Loads, Not Bequests: An Order-Independent Decomposition of the Annuity Puzzle*

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July 11, 2026

Abstract

Why is voluntary annuity ownership only a few percent when Yaari (1965) implies full annuitization? Competing explanations abound, but the sequential decompositions that weigh them are order-dependent. We nest nine rational and preference channels in a calibrated lifecycle model of single US retirees and attribute the demand gap below the frictionless benchmark with exact Shapley values over all $2^9 = 512$ subsets. Pricing loads dominate (30.9 pp); survival pessimism (12.4 pp) is the sole second-tier suppressor; the correlated health–mortality channel's attribution is near zero (0.8 pp) once aggregate mortality is anchored, though its grand-coalition removal still collapses ownership. Bequest motives, the most-cited explanation, are mid-pack (3.9 pp); pre-existing annuitized income raises demand on average rather than suppressing it. The baseline model overpredicts the two untargeted Health and Retirement Study ownership measures severalfold (19.3% against 3.64% and 6.06%), and the level is fragile, swinging from 0.0% to 28.2% across the plausible risk-aversion range. The contribution is the order-independent attribution and the stability of the ranking, not resolution of the level: pricing loads lead throughout that range, in 3 of the four wealth bands (a near-tie with pessimism in the top band), and at alternative baseline pricing. A policy application: a projected 22% Social Security benefit cut raises private annuitization

*Replication code and data are available at `https://github.com/DerekTharp/annuity-puzzle-model`.

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Declaration of interest: the author owns a financial advisory business whose revenue is predominantly asset-under-management fees, and is licensed to sell income annuities. This research received no external funding. The HRS (Health and Retirement Study) is sponsored by the National Institute on Aging (grant number NIA U01AG009740) and is conducted by the University of Michigan; this paper uses the RAND HRS Longitudinal File 2022 (V1) (Health and Retirement Study, 2025).

only among pension-linked upper-middle and wealthy households. In a two-product extension with coverage-calibrated access to group pricing, the bottom half of the wealth distribution does not respond and lower-middle participation falls, so neither retail nor group markets backfill the cut where it bites hardest.

JEL Codes: D14, D15, D91, G22, H55, J14
Keywords: annuity puzzle, lifecycle model, Shapley decomposition, stochastic mortality, age-varying consumption needs

1 Introduction

Yaari (1965) established one of the sharpest predictions in consumer theory: an individual facing uncertain lifetime, with no bequest motive and access to actuarially fair annuities, should convert all wealth into a life annuity. The mortality credit, the return premium financed by redistributing wealth from decedents to survivors, dominates any asset with the same underlying return. Davidoff et al. (2005) extended this result to incomplete markets, showing that partial annuitization remains optimal under a wide range of conditions. Yet observed voluntary annuity ownership among single US retirees aged 65–69 is only a few percent: 3.64% on the cleaner HRS lifetime annuity contract indicator (q286 series) and 6.06% on the conventional any-annuity income proxy used in prior literature (Lockwood, 2012; Dushi and Webb, 2004; Poterba et al., 2011).[1] A companion survey (Tharp, 2026a) reads the evidence conditionally: the accumulated channels well explain low demand for many household types but leave the welfare question contested for high-wealth, single, childless retirees in good health, and it identifies the absence of a unified structural model incorporating all empirically relevant channels as the gap preventing resolution of that contested cell. This paper supplies the closest such model to date; the principal omissions (housing illiquidity and intra-household risk sharing) both work to reduce demand further (Section 6).

Six decades of research proposed partial explanations for this gap (recently surveyed in Hershfield et al., 2026), and the standard tool for adjudicating among them, the sequential decomposition (channels are switched on one at a time and the drop in predicted ownership attributed to each), returns an answer that depends on the order of entry. Lockwood (2012) emphasized bequest motives; Reichling and Smetters (2015) the correlation between health shocks and mortality; Mitchell et al. (1999) and others pricing loads. Because the channels interact, the ranking is path-dependent: the same model can crown different dominant frictions depending on which channel enters late, when little surplus remains to attribute. The disagreement in the literature was, in part, an artifact of the decomposition method.

This paper nests nine rational and preference channels in a single calibrated lifecycle model (pre-existing Social Security annuitization, bequest motives, combined medical-expenditure risk and the Reichling and Smetters (2015) health-mortality correlation, subjective survival pessimism (O'Dea and Sturrock, 2023), pricing loads, inflation erosion, public-care aversion (Ameriks et al., 2011, 2020), age-varying consumption needs (Aguiar and Hurst, 2013), and state-dependent utility (Finkelstein et al., 2013)) and attributes the annuity-

[1]The two HRS measures differ because the income proxy (*rwiann*) classifies any positive annuity income as ownership, including one-time DC pension withdrawals and short-period payouts; the lifetime contract indicator (the question stem "Will this continue for the rest of your life?") isolates true life-contingent annuity contracts. Section 3 discusses both measures.

demand gap with exact Shapley values computed over all $2^9 = 512$ channel subsets. The Shapley value averages each channel's marginal contribution across every entry order, so the attribution is order-independent by construction. Where a sequential decomposition reports one of many possible rankings, the Shapley reports a unique attribution given the channel partition, outcome statistic, and calibration. The result is an accounting decomposition within one calibrated model, a ranking of which channels suppress demand most in this framework, rather than a quasi-experimental identification of causal magnitudes across the literature.

Within this game, the ranking is clear. Pricing loads are the dominant suppressor, contributing 30.9 percentage points of the total demand reduction. Subjective survival pessimism (12.4 pp) is the sole second-tier suppressor; the combined medical and Reichling and Smetters (2015) health-mortality channel attributes near zero (0.8 pp) once aggregate mortality is anchored, though removing it from the grand coalition still collapses ownership. Bequest motives, the single most-cited explanation for the puzzle, contribute 3.9 pp, mid-pack. Social Security carries a large negative Shapley value (-26.2 pp): averaged across channel configurations it is the largest-magnitude demand-boosting force, because the public income floor it provides is what makes a private annuity feasible. Yet at the fully-loaded margin the benefit-cut counterfactual perturbs, the same income substitutes for private annuities (Section 4). Most single retirees are substantially annuitized through Social Security before they face a private annuity decision, so the marginal value of additional annuitization is small, and the suppressing channels need not be individually large because they compound on a thin remaining surplus.

The ranking is stable where the predicted level is not. The level is knife-edged in risk aversion: predicted ownership moves from 0.0% to 28.2% as γ ranges over $[2.0, 3.0]$, because the extensive margin sits on a fixed-cost threshold that the population crosses discontinuously (predicted ownership is invariant as the minimum-purchase floor rises from \$0 to \$25,000, since buyers are top-band households purchasing well above any such floor). The same fragility plausibly explains why broadly comparable models have predicted ownership anywhere from a few percent to full annuitization. The top of the channel ranking, by contrast, is stable: pricing loads remain the dominant suppressor on the ownership statistic at all 5 values of γ in the $[2.0, 3.0]$ stability set, in 3 of the four wealth bands, when the game is recomputed at alternative baseline money's worth ratios of 0.85 and 0.90, at finer solution grids than production, and when the two multiplicative-utility channels are removed from the game as players (Section 5). The stability set has boundaries, which we report rather than hide: at $\gamma = 2.0$, where the participation margin is degenerate, survival pessimism leads the continuous mean-purchase statistic, and bequest motives take the largest attribution on both statistics only at $\gamma = 1.5$, below the set (suppressor rank correlation across the two statistics is 0.96 at the production calibration). The lower tiers are less rigid: survival pessimism is second at every γ in the set, while the third slot rotates from bequest motives at low risk aversion to age-varying needs at the baseline, with the Med+R-S channel entering the top three only at $\gamma = 3.0$. In calibrated annuity models, then, the predicted participation *level* is a poor basis for claims; the dominant-channel result and the qualitative comparative statics are the stable findings the paper relies on.

Social Security is the annuity almost every US retiree holds, and for most households the only one. When the Old-Age and Survivors Insurance trust fund depletes in late 2032, current projections imply an across-the-board benefit cut of roughly 22% absent legislative action (Board of Trustees, Federal Old-Age and Survivors Insurance and Federal Disability Insurance Trust Funds, 2026). A natural public-finance question is whether private annuity markets could absorb the longevity risk a smaller public benefit would leave exposed. The answer in this paper, for the retail immediate annuity (the one longevity product an individual household can buy on its own), is that they would not, and that the failure is distributional: a benefit cut raises private annuity demand mainly in the top wealth band, with a smaller response in the upper-middle band, a slight decline in the lower-middle band, and none at the bottom, because the households the cut most exposes do not, and at current prices rationally would not, purchase private annuities. Reaching that answer requires first explaining why voluntary annuity demand is so low and resolving a methodological disagreement about what suppresses it.

Resolving the attribution reframes the puzzle as distributional. The proportional pricing load is the binding margin across the wealth distribution: at the baseline money's-worth ratio, immediate annuitization is value-destroying for all but the wealthiest, so predicted ownership rises with wealth and concentrates at the top (80.9% in the top band, near zero below). A household-level diagnostic confirms the mechanism: removing the proportional load lifts the would-annuitize share in the upper-middle wealth band from under a quarter to near-universal (Section 4.9). Observed lifetime-annuity ownership in the HRS also rises with wealth but is interior in the middle bands, where the single-product model predicts none; we attribute this gap to better-priced employer and group annuities outside the model and to preference heterogeneity the representative-band agent omits, and report it rather than fit it. The policy counterfactuals inherit this structure. A 22% Social Security cut, approximating the projected OASI shortfall when the trust fund depletes in late 2032 (Board of Trustees, Federal Old-Age and Survivors Insurance and Federal Disability Insurance Trust Funds, 2026), raises predicted private annuity demand, but the response is concentrated among the wealthy: top-band ownership rises from 80.9% to 95.9%, while the bottom band, the most exposed to the cut, does not respond (0.0%). Retail longevity insurance does not backfill a public benefit cut where the cut bites hardest.

The cross-sectional implications of the ranking are testable in the same data used to discipline the model. In pooled HRS samples of single retirees aged 65–69, the point estimates carry the predicted signs for the resource and health channels: ownership rises with wealth, falls with poor health, and falls with pre-existing annuitized income conditional on wealth (5 of 7 channel-implied signs match). They are individually imprecise in this modest sample, so the check is directional rather than sharp. The two sign mismatches come from the expectation-based channels (subjective survival and bequest intention), whose survey proxies are confounded in ways the structural calibration is designed to bypass.

Relative to prior multi-channel papers, the present model broadens the channel set and changes the attribution. Pashchenko (2013) jointly considered Social Security, bequests, minimum purchase requirements, and pricing frictions, but not the Reichling–Smetters health-survival mechanism, subjective survival pessimism, or age-varying needs. Peijnenburg et al. (2016) emphasized incomplete markets and background risk, concluding that the annuity puzzle remains unresolved in their framework. This paper shows that pricing and late-life expenditure decline are quantitatively central once evaluated in a common model, while correlated health and mortality risk is central in the sequential path and on the grand-coalition margin but near zero in the order-independent baseline attribution.

The welfare results follow the same distributional logic. Under current pricing, annuity-market access is approximately welfare-neutral across the wealth distribution, because most calibrated households optimally do not annuitize. Larger gains come from interventions that change the optimal choice: group annuity pricing (MWR = 0.90) on the supply side, and default architecture on the demand side, drawing on the broader evidence that defaults strongly shape retirement-saving choices (Madrian and Shea, 2001). The two operate on separate margins.

The paper makes three contributions. First, it shows that order-dependence can by itself generate contradictory channel rankings within a single fixed model (bequest motives earn a 0.0 pp attribution entering second in one sequential ordering, 7.7 pp as a grand-coalition removal, and 3.9 pp as their order-free Shapley value), so the literature's disagreement over the dominant channel is at least partly an artifact of the tool. It then delivers an order-independent attribution in which pricing loads are the dominant suppressor of voluntary annuity demand across the $\gamma \in [2.0, 3.0]$ stability set and at alternative baseline money's worth ratios, with survival pessimism second at every γ in the set, a third slot that rotates (bequest motives at low risk aversion, age-varying needs at the baseline, Med+R-S only at $\gamma = 3.0$), and bequest motives (the literature's leading explanation) mid-pack on the ownership margin at the baseline. Second, it establishes the level/ranking distinction as a methodological discipline for calibrated annuity models: the predicted participation level is fragile across risk aversion and grid resolution, while the ranking and the policy comparative statics are stable. Third, it reframes the annuity puzzle from an aggregate behavioral anomaly into a distributional question: the proportional pricing load is the margin that excludes the non-wealthy from retail immediate annuitization, and by-band counterfactuals show that even group (MWR = 0.90) or public-option (MWR = 0.95) pricing draws in mainly the top wealth band. A two-product extension with coverage-calibrated access to institutional pricing generates interior ownership and the observed low-to-middle gradient the single-product model cannot, while overshooting the third band and the top band, and re-running the benefit cut inside it yields a two-signed incidence: annuitization rises among pension-linked upper-middle and wealthy households but falls in the lower-middle band, where the cut's income effect removes the floor that made the group annuity rational. The incidence result is a comparative static of the full model whose top-heavy concentration is robust across treatments of the participation margin (Section 4.9). It does not rest on the Shapley attribution. (The theta-dispersed treatment distributes bequest heterogeneity using the self-reported bequest-intention covariate whose sign the cross-sectional logit mispredicts; the use here is dispersion-only, and the robustness conclusion does not rest on it.) Whether employer- or group-sourced annuities (the products the observed interior ownership points to, which differ from retail SPIAs in more than price) could reach the households a cut most exposes is the open question the single-product model leaves for future work. Beyond these three structural contributions, an exploratory extension flags an agenda. Structural annuity decompositions almost never include behavioral channels; when two are added at magnitudes drawn from the behavioral literature, source-dependent utility alone moves predicted ownership by more than any structural channel's Shapley attribution, and the narrow-framing purchase penalty saturates the participation margin outright, collapsing ownership to essentially zero. The exercise neither identifies nor ranks them; its point is that an almost-unexplored behavioral frontier could outweigh the rational and preference forces the literature studies, a reason for caution about any structural-only attribution, this one included. The structural ranking is computed with these channels held off.

These results shift the question from whether retirees irrationally fail to annuitize to a more policy-relevant one: which channels suppress annuity demand most, and which market reforms would make annuitization welfare-improving for particular households?

Section 2 specifies the model. Section 3 describes the calibration. Section 4 presents the decomposition, counterfactuals, and welfare analysis. Section 5 reports sensitivity checks. Section 6 discusses limitations. Section 7 concludes.

2 Model

2.1 Environment

Time is discrete. An individual enters at age 65 (period $t = 1$) and may survive to a maximum age of 110 (period $T = 46$). At entry, the individual makes a single irreversible decision: what fraction $\alpha \in [0, 1]$ of initial liquid wealth W_0 to convert into a single premium immediate annuity (SPIA). The remaining wealth $(1 - \alpha)W_0$ stays liquid. In each subsequent period, the individual chooses consumption c_t from available resources. The backward-induction solution yields a value function and annuitization policy at every age, not only age 65. When the model is taken to the HRS sample of single retirees aged 65–69 (Section 3), each respondent's one-shot annuitization decision is evaluated at their *observed* age, using the age-appropriate value function and a SPIA priced at that age. The age-65 entry point describes the lifecycle problem the value function solves; sampled retirees need not decide at exactly 65. Forcing the entire sample to age 65 lowers predicted ownership from 21.2% to 17.2% at the 60×20 evaluation grid (a level shift of the same character as the calibration sensitivities in Section 5) while preserving the features of the channel ranking on which the paper's claims rest: recomputing the structural Shapley with every respondent at age 65, pricing loads remain the dominant suppressor and Social Security the largest demand booster, with bequests still mid-pack.

The state at period t has four variables: liquid wealth $W_t \geq 0$, annuity income A (fixed after the age-65 purchase), health status $H_t \in \{1, 2, 3\}$ (Good, Fair, Poor), and age t. Social Security income $\mathrm{SS}(t)$ arrives exogenously each period.

2.2 Preferences

The individual maximizes expected discounted lifetime utility. Flow utility over consumption takes the constant relative risk aversion (CRRA) form:

$$u(c) = \frac{c^{1-\gamma}}{1-\gamma}, \quad \gamma > 0, \quad \gamma \neq 1. \tag{1}$$

The coefficient of relative risk aversion γ governs both risk aversion and the elasticity of intertemporal substitution.

Bequest utility follows the De Nardi–French–Jones specification, which nests bequests as a luxury good:

$$v(b) = \theta \frac{(b + \kappa)^{1-\gamma}}{1-\gamma}, \tag{2}$$

where b is bequeathable wealth (liquid wealth at death; annuity income ceases), $\theta \geq 0$ controls bequest intensity, and $\kappa > 0$ governs how strongly bequests behave as a luxury good (De Nardi, 2004). When κ is large relative to typical wealth, marginal bequest utility is nearly flat for low-wealth individuals and declines steeply for the wealthy. Lockwood (2012) estimated this specification on Health and Retirement Study (HRS) data; it captures the concentration of bequest motives among high-wealth households.

The parameters θ and κ must be interpreted jointly. With the DFJ luxury-good calibration ($\theta = 56.96$, $\kappa = \$272{,}628$), an individual with \$50,000 in wealth faces negligible marginal bequest utility, while an individual with \$1,000,000 faces strong resistance to annuitizing. Combining this θ with a small κ (e.g., \$10, the homothetic specification) produces pathological results. Extreme marginal bequest utility near $b = 0$ acts as a precautionary buffer rather than a genuine bequest motive and anomalously *increases* predicted ownership above the no-bequest case.[2] The DFJ estimates from Lockwood (2012) are jointly calibrated to HRS bequest data; using one parameter with the other replaced is not a valid counterfactual.

The bequest channel's isolated effect on ownership is modest under the DFJ specification (100% retention rate in the sequential decomposition) because the luxury-good κ makes the bequest motive nearly irrelevant for the median retiree in the HRS sample. The 100% retention rate reflects the ordering: bequests are added when ownership is already 100% (steps 0–1 include only SS). In the full nine-channel structural model, removing bequests raises ownership from 19.3% to 27.0% (Table A9), because bequests interact with pricing loads and inflation to suppress demand for agents near the annuitization margin. The Shapley decomposition, which averages over all orderings, assigns bequests a value of 3.9 pp (15% share).

The individual discounts future utility at rate β per period.

2.3 Health dynamics and mortality

Health follows a first-order Markov process with three states: Good ($H = 1$), Fair ($H = 2$), and Poor ($H = 3$). Transition probabilities are age-dependent and calibrated to HRS panel data following De Nardi et al. (2010):

$$\Pr(H_{t+1} = h' \mid H_t = h, \text{age} = t) = \pi(h, h', t). \tag{3}$$

[2] At $\kappa = \$10$, marginal bequest utility near $b = 0$ is $\theta \cdot 10^{-\gamma}$, which is extreme relative to marginal consumption utility. Individuals avoid annuitizing not to protect bequests, but to avoid near-zero-wealth states where $v'(b)$ becomes very large. The implementation applies a small numerical floor at $\max(b + \kappa, 1)$ to prevent overflow at $b = 0$ when $\kappa = 0$ is used as a robustness check; with the production DFJ calibration ($\kappa = \$272{,}628$) the floor never binds. See Appendix C for details.

Mortality depends on health through a proportional hazard specification. Baseline survival probabilities $s_{\text{base}}(t)$ come from a sex-blended SSA 2003 period life table matched to the sample's sex composition (Section 3). Health adjusts survival through:

$$s(t, H) = s_{\text{base}}(t)^{\mu(H,t)}, \tag{4}$$

where $\mu(H, t)$ is a health- and age-specific hazard multiplier with $\mu(\text{Good}, t) < 1 < \mu(\text{Poor}, t)$ and $\mu(\text{Fair}, t) = 1$. This specification maps directly to proportional hazards: if the baseline hazard is $\lambda_{\text{base}}(t) = -\ln s_{\text{base}}(t)$, then the adjusted hazard is $\mu(H, t) \cdot \lambda_{\text{base}}(t)$. In production, a common age-specific factor rescales the multipliers so the population-weighted survival reproduces $s_{\text{base}}(t)$ exactly at every age (health shares evolve from the sample's pooled ages 65–69 distribution under the transition matrices); hazard ratios across health states are unchanged, and without the anchor the health mixture would overstate aggregate mortality relative to the life table.

The baseline specification uses constant hazard multipliers $\mu = [0.50, 1.0, 3.75]$ for Good, Fair, and Poor health, set above the HRS self-reported age-band estimates toward the steeper functional-limitation gradient of Reichling and Smetters (2015). The health-mortality gradient compresses with age: the Poor-to-Fair hazard ratio falls from 3.29 at ages 65–74 to 2.77 at 75–84 to 1.82 at 85+. Section 5 reports results across the empirical range of multiplier specifications.

The age-varying multipliers are the mechanism underlying the Reichling and Smetters (2015) result. A negative health shock simultaneously increases mortality ($\mu(\text{Poor}, t) > 1$ reduces expected remaining annuity payments) and, through the medical expenditure process described below, increases the need for liquid wealth. The annuity becomes "valuation-risky," losing value in precisely the states where liquidity is most valuable. The effect is strongest at younger ages, where the gradient is widest and the age-65 annuitization decision is made.

2.4 Subjective survival beliefs

O'Dea and Sturrock (2023) document that individuals aged 65–69 underestimate their survival probability at age 75 by approximately 15 percentage points relative to actuarial tables (subjective Pr(survive to 75) ≈ 0.71 versus actuarial 0.86). Pessimism reduces the perceived value of annuities by overweighting the probability of dying before recovering the premium.

The model allows the agent's subjective survival probabilities to differ from the objective rates used by the insurer to price annuities:

$$s^{\text{subj}}(t, H) = \psi \cdot s(t, H), \quad \psi \in (0, 1]. \tag{5}$$

The parameter ψ scales each one-year survival probability downward. The insurer prices the annuity from the health-independent population life table; only the agent's Bellman equation uses the health-conditioned, pessimism-scaled s^{subj}. The baseline value is the paper's focal empirical translation: O'Dea and Sturrock (2023) provide the cleanest age-matched anchor, and their 65–69 ten-year survival ratio implies a per-year factor $\psi = (0.71/0.86)^{1/10} \approx 0.981$. The baseline adopts this value because it is the direct translation of the age-matched evidence, not an endpoint chosen for fit. Survival miscalibration is real but its sign is age-dependent. Heimer et al. (2019) document a crossover near ages 65–70 (younger individuals underestimate and older individuals overestimate survival), so mild optimism (ψ at or above one) is plausible for this cohort. We therefore report ψ over the range $[0.96, 1.0]$, spanning a strong-pessimism endpoint, the O'Dea–Sturrock baseline, and objective beliefs, with predicted ownership across it in Section 5. The participation level is sensitive to ψ, consistent with the paper's position that the level is not the contribution; the paper's claims are built to be level-agnostic. Pessimism's own attribution is also ψ-dependent: recomputing the game at the strong-pessimism endpoint $\psi = 0.96$ leaves the ranking intact, with pessimism's contribution larger there, as expected, and it remains the second-largest suppressor at both the baseline and the endpoint; the features the paper's claims rest on (pricing loads dominant, bequests mid-pack) hold at both (Appendix A.5). When $\psi = 1$, beliefs are objective and this channel is inactive.

2.5 Medical expenditures

Out-of-pocket medical expenditures depend on age and health. Log medical spending follows:

$$\ln m_t = \mu_m(t, H_t) + \sigma_m(t, H_t) \cdot \varepsilon_t, \quad \varepsilon_t \sim N(0, 1), \tag{6}$$

where μ_m and σ_m are the mean and standard deviation of log expenditures, both age- and health-dependent. The baseline calibration matches the age profiles reported by Jones et al. (2018): mean out-of-pocket spending of \$4,201 at age 70, rising to a combined out-of-pocket-plus-Medicaid mean of \$29,703 at age 100, with a combined 95th percentile of approximately \$111,509 at age 100 (the model's expenditure draw is gross of the Medicaid transfer, so the combined moments are the appropriate old-age targets).

A Medicaid safety net provides a consumption floor $\underline{c}$. If total resources (wealth plus income minus medical costs) fall below $\underline{c}$, the government covers the shortfall. The individual consumes $\underline{c}$ and enters the next period with zero liquid wealth. The floor is \$6,180, following Lockwood (2012), who chose it to approximate the SSI federal benefit rate plus average state supplement for an elderly individual at the time of his study; the value is carried over at its original nominal level rather than restated in 2014 dollars.

Nine-node Gauss–Hermite quadrature computes expectations over medical expenditure shocks. At the calibrated variance ($\sigma \approx 1.4$), lower-order rules place insufficient weight in the tails of the lognormal distribution; Appendix G documents the convergence diagnostics.

2.6 Annuity pricing

The annuity is nominal: the insurer prices the stream of nominal payments using the nominal discount rate $r_{\text{nom}} = (1+r)(1+\pi)-1$, where r is the real interest rate and π is the inflation rate. The actuarially fair annual payout per dollar of premium is:

$$\text{payout}_{\text{fair}} = \frac{1}{\sum_{k=0}^{T-1} \frac{\prod_{j=0}^{k-1} s_{\text{base}}(j)}{(1+r_{\text{nom}})^k}}. \tag{7}$$

The nominal discount rate reflects the insurer's investment in nominal bonds. The resulting initial payout is higher than under real discounting, but the consumer's real income from the annuity declines each period:

$$A_t^{\text{real}} = \frac{A_1}{(1+\pi)^{t-1}}, \tag{8}$$

where π is the annual inflation rate. At $\pi = 2\%$, purchasing power falls by 39% over 25 years.

When $\pi = 0$ (the Yaari benchmark and intermediate decomposition steps where inflation is inactive), $r_{\text{nom}} = r$ and the pricing reduces to the standard real-annuity formula.

The annuity is priced with a money's worth ratio (MWR), defined as the expected present value of payouts per dollar of premium. The loaded payout rate is MWR $\times$ $\text{payout}_{\text{fair}}$. We set MWR = 0.87 at baseline, consistent with the population-table estimates of Mitchell et al. (1999) for the US individual annuity market. Mitchell et al. (1999) estimated MWRs of 0.80–0.85 using population mortality tables, the relevant benchmark for an individual considering annuitization (the "money's worth" of the purchase against population life expectancy). Using annuitant mortality tables, which adjust for selection, yields higher MWRs of 0.90–0.94, but this overstates the value proposition for the marginal non-annuitant. More recent estimates from Wettstein et al. (2021) put population-table MWRs around 0.85. The production value of 0.87 sits slightly above both cited anchors and is conservative for the loads-dominate conclusion, which strengthens at lower MWR. Section 5 reports full sensitivity: at MWR = 0.85, predicted ownership falls to 15.4%; at MWR = 0.90, to 24.3%. The steep sensitivity to MWR is itself a finding: it identifies pricing as the most actionable policy lever.

A fixed cost of \$2,500 is incurred upon any purchase, and annuitization requires a minimum premium of \$10,000 (an insurer underwriting minimum that turns out to be non-binding; Section 5).[3]

2.7 Budget constraint

Within each period, the timing is:

1. The individual enters with liquid wealth W_t, annuity income A_t, Social Security $\mathrm{SS}(t)$, and health H_t.
2. Medical expenditure m_t is realized.
3. The individual chooses consumption c_t.
4. Remaining wealth earns the risk-free return r.
5. Health transitions to H_{t+1}.
6. The individual survives to $t+1$ with probability $s(t, H_t)$.

The intertemporal budget constraint is:

$$W_{t+1} = (1 + r)(W_t + A_t + \mathrm{SS}(t) - m_t - c_t), \tag{9}$$

subject to $c_t \geq \underline{c}$ and $W_{t+1} \geq 0$ (no borrowing).

2.8 Age-varying consumption needs

Aguiar and Hurst (2013) document that consumption expenditure declines at roughly 2% per year in retirement, even among wealthy households, once time costs and home production are accounted for. The decline reflects changing consumption needs (reduced transportation, clothing, and food-away-from-home expenditures) rather than binding liquidity constraints, and it reduces the value of late-life income streams, including annuities.

The model captures this through an age-dependent weight on flow utility:

$$w(t) = (1 - \delta_c)^{t-1}, \quad \delta_c = 0.02, \tag{10}$$

[3] The MWR is computed using beginning-of-period payment timing: the first payment occurs at purchase. This convention is standard in the annuity pricing literature (Mitchell et al., 1999) and matches the implementation in the replication code.

where $t = 1$ at age 65. Flow utility becomes $w(t) \cdot u(c)$, an age-dependent felicity shifter: as consumption needs decline, the marginal value of late-life consumption falls, reducing the insurance value of a level real income stream. When $\delta_c = 0$, the channel is inactive and the model reduces to the standard time-separable case.

Age-varying consumption needs differ from time preference (β). The discount factor governs how future utility is discounted; $w(t)$ governs how much utility a given level of consumption generates at each age. A constant β with declining $w(t)$ implies the individual values future consumption opportunities but needs less consumption for the same felicity as she ages.

2.9 State-dependent utility

Finkelstein et al. (2013) estimate that individuals in poor health value a dollar of consumption roughly 10–25% less than those in good health. Reichling and Smetters (2015) incorporated this channel in their annuity model. State-dependent utility reduces annuity demand because the annuity delivers income in poor-health states where its marginal utility is lower.

The model allows the marginal utility of consumption to depend on health status through a multiplicative weight:

$$u(c, H) = \varphi(H) \cdot \frac{c^{1-\gamma}}{1-\gamma}, \tag{11}$$

where $\varphi(\text{Good}) = 1.00$, $\varphi(\text{Fair}) = 0.92$, and $\varphi(\text{Poor}) = 0.82$, the central midpoint of the Finkelstein et al. (2013) estimates, applied through the multiplicative state-dependent-utility approach of Reichling and Smetters (2015). When $\varphi(H) = 1$ for all H, this channel is inactive.[4]

In practice, state-dependent utility has a negligible effect on predicted ownership (Shapley value of 0.3 pp). The channel is included to confirm that its quantitative irrelevance, suggested by the prior literature, holds in the unified framework.

[4]The age and health weights $w(t)$ and $\varphi(H)$ enter multiplicatively on CRRA flow utility. For $\gamma > 1$, $u(c) < 0$, so a weight below one raises the utility *level* while lowering marginal utility $\varphi(H)\, w(t)\, c^{-\gamma}$; choices are driven by marginal utility, so the behavioral direction (poor health and older age lower the value of a level income stream) is correct. Because the weights are homogeneous of degree $1 - \gamma$ in consumption, they cancel in within-state comparisons of consumption-financed utility. Bequest utility is unscaled, and the compensating-variation computation handles it through the exact additive bequest decomposition of Section 4.11 rather than by cancellation, so the reported CEVs do not rest on a blanket invariance claim. Public-care aversion (χ_{LTC}), by contrast, is identified from a cross-state comparison (Medicaid- versus self-financed care) and is therefore applied as a consumption-equivalent transformation, which carries the correct sign under any γ. The transformation acts only in states where the agent is simultaneously Medicaid-reliant and in poor health; in those states realized consumption is pinned at the public floor, so the correction adjusts the realized flow utility without altering the already-constrained consumption choice, while the precautionary incentive to avoid such states enters earlier decisions through the backward-induction continuation value.

2.10 Behavioral channels (exploratory)

The nine rational and preference-based channels above constitute the structural baseline. Two further behavioral channels are exploratory extensions layered onto that baseline.

Source-dependent utility (SDU; exploratory). Blanchett and Finke (2024, 2025) document that retirees apply a higher consumption rate to guaranteed income flows than to portfolio assets, consistent with the source-dependent utility framework of Shefrin and Thaler (1988) and Thaler (1999). The model captures this through a multiplicative discount on portfolio drawdowns:

$$c_{\text{eff}}(t) = c_{\text{income}}(t) + \lambda_W \cdot c_{\text{portfolio}}(t), \quad \lambda_W \in (0, 1], \tag{12}$$

where c_{income} is consumption financed from annuity and Social Security income, $c_{\text{portfolio}}$ is consumption financed from liquid wealth drawdowns, and $\lambda_W < 1$ reflects the lower utility weight placed on portfolio-financed consumption. The production calibration sets $\lambda_W = 0.625$ as a literature-magnitude best guess: Blanchett and Finke (2024) report retirees spending at approximately 50% of the rate from portfolio assets compared with 80% from guaranteed income, implying $\lambda_W = 50/80 = 0.625$. The parameter is anchored to a behavioral spending differential, not moment-matched to a US annuitization moment; results across $\lambda_W \in \{0.5, 0.625, 0.75, 0.85\}$ are reported in Section 5.

Narrow-framing at-purchase penalty (PED; exploratory). A cumulative-prospect-theory mechanism formalized by Hu and Scott (2007), with empirical support from Brown et al. (2008) and Chalmers and Reuter (2012), generates a per-period at-purchase disutility while the household's running gain/loss tally on the SPIA is underwater. Letting $\pi = \alpha W_0$ denote the premium, $A = \pi \cdot \rho$ the annual annuity income at payout rate ρ, and $t^* = \lceil 1/\rho \rceil + 1$ the breakeven horizon, the model captures the penalty as the discounted, survival-weighted NPV of the loss-aversion stream:

$$\Delta V_{\text{purchase}} = -\psi_{\text{purchase}} \cdot u'(c_{\text{ref}}) \cdot \sum_{t=1}^{t^*} \beta^{t-1} \cdot S(t) \cdot \max(0, \pi - A(t-1)) \cdot \mathbf{1}(\alpha > 0), \tag{13}$$

where $c_{\text{ref}} = \$18{,}000$ is a reference consumption level chosen so that ψ_{purchase} has an interpretable scale, $u'(c_{\text{ref}}) = c_{\text{ref}}^{-\gamma}$ is the marginal utility of consumption at c_{ref} (converting dollar-denominated underwater amounts to utility units), $S(t)$ is the cumulative survival probability to period t, and β is the discount factor. The penalty applies once at age 65 over the underwater period of the SPIA; once cumulative payouts cross the premium (period t^*) the loss tally turns positive and the penalty vanishes. One convention is worth making explicit: $S(t)$ is the objective population survival that prices the contract, not the health-conditioned subjective survival used in the continuation value. The underwater tally is a property of the contract horizon, and weighting the penalty by pessimistic beliefs would double-count the belief distortion the pessimism channel already carries. The production calibration sets $\psi_{\text{purchase}} = 0.05$ as a literature-magnitude best guess, not moment-matched to a US annuitization target; results across $\psi_{\text{purchase}} \in \{0.01, 0.05, 0.09\}$ are reported in Section 5. At the production magnitude the at-purchase penalty saturates (it eliminates participation for nearly all agents), so the eleven-channel specification illustrates where literature-magnitude behavioral parameters would push predictions rather than serving as a moment-matched calibration.

Status of behavioral channels. Both SDU and PED are exploratory rather than identified. The model could be moment-matched against a US annuitization target, but doing so would conflict with the out-of-sample comparison strategy used elsewhere in the paper (the structural baseline uses no US annuitization moment). The eleven-channel specification therefore shows what happens when these channels enter at literature magnitudes, as a sensitivity exercise rather than a within-model calibration of the behavioral primitives.

2.11 Bellman equation

For $t < T$, the value function satisfies:

$$\begin{aligned} V(W, A, H, t) = \mathbb{E}_m \Big[\max_c \Big\{ & w(t) \cdot u(c, H) \\ & + \beta\, s(t, H)\, \mathbb{E}_{H'} \Big[V(W', A', H', t+1) \mid H \Big] \\ & + \beta\, (1 - s(t, H))\, v(W') \Big\} \Big], \end{aligned} \tag{14}$$

where the agent observes the medical shock m at the start of the period, so the maximization over c is conditional on m and the outer expectation is over the medical-shock distribution. Medical expenses absorb the period's income flows before drawing down portfolio wealth; this convention matters only when the source-dependent utility channel is active, since it determines how much residual consumption each source finances. The health transition ($H \to H'$) occurs after the consumption decision and enters as an inner expectation in the continuation. $W' = (1 + r)(W + A_t + \mathrm{SS}(t) - m - c)$, $A' = A/(1 + \pi)$ is next period's real annuity income, $w(t)$ is the age-varying needs weight from equation (10), and $u(c, H)$ is the state-dependent utility from equation (11). When both channels are inactive ($\delta_c = 0$,

$\varphi(H) = 1$), this reduces to the standard Bellman equation. At the terminal period T, survival probability is zero and the individual divides available resources between final consumption and a terminal bequest, consuming everything when the bequest motive is inactive.

At $t = 1$ (age 65), the annuitization decision is:

$$\alpha^* = \arg\max_{\alpha\in[0,1]} V\Big((1-\alpha)W_0 - F\cdot \mathbf{1}(\alpha > 0),\ \alpha W_0 \cdot \text{payout},\ H_0,\ 1\Big), \tag{15}$$

where F is the fixed purchase cost.

2.12 Analytical characterization

The numerical decomposition in Section 4 has analytical structure. At $\alpha = 0$, shifting one dollar from liquid wealth to the annuity premium buys p dollars of annual annuity income, where p is the payout rate, and changes welfare by

$$\left.\frac{dV}{d\alpha}\right|_{\alpha=0} = W_0\Big[p\cdot V_A(W_0, 0, H, 1) - V_W(W_0, 0, H, 1)\Big], \tag{16}$$

where V_W and V_A are the partial derivatives of the value function with respect to liquid wealth and annuity income. The payout rate embeds the pricing load: $p = \text{MWR}/a(r)$, where $a(r) = \sum_t S(t)/(1+r)^t$ is the actuarially fair annuity factor, so a fairly priced annuity (MWR $= 1$) pays $p = 1/a(r)$. Define the *annuity value ratio*

$$R \equiv p\cdot\frac{V_A}{V_W} = \text{MWR}\cdot\frac{V_A}{a(r)\,V_W}. \tag{17}$$

The agent annuitizes a positive fraction if and only if $R > 1$.

Proposition 1 (Yaari)**.** *With no bequests ($\theta = 0$), deterministic health, fair annuity pricing, and $\beta(1+r) = 1$, the envelope theorem implies $V_W = u'(c^*)$ at the optimum. The annuity-income derivative V_A is the present discounted value of the future marginal utilities generated by the income stream, weighted by survival probabilities; with consumption smoothed across dates and states, $V_A = a(r)\,u'(c^*)$. The annuity factor cancels against the fair payout rate $p = 1/a(r)$, yielding $R = 1$ with full annuitization weakly optimal.*

Each demand-suppressing channel introduces a factor $\Phi_i < 1$ that scales R below the Yaari benchmark. Appendix I develops this multiplicative factorization and the super-additive participation logic it implies; both are heuristic guides rather than proven identities, and the exact, model-consistent attribution is the Shapley decomposition of Section 4.4. Two terms anchor intuition. The pricing wedge scales R one-for-one ($\Phi_{\text{MWR}} = \text{MWR} = 0.87$), and under the DFJ luxury-good specification the bequest factor is near one for median-wealth retirees, the analytical seed of the loads-dominant, bequests-mid-pack ranking.

2.13 Solution method

The model is solved by backward induction over the state space (W, A, H, t). The wealth grid uses 80 points with power-function spacing (denser at low wealth) over $[0, \$3 \text{ million}]$, covering the 99.5th percentile of the HRS wealth distribution. The annuity income grid uses 30 points with cubic spacing to resolve small purchases, and health is discrete with three states. Brent's method optimizes consumption at each grid point; the wealth dimension uses piecewise linear interpolation with flat extrapolation.

The annuitization fraction α is optimized over a 101-point grid at age 65. Grid convergence is checked at the 9-node production quadrature rule under the production per-quartile Social Security assignment: refinement near the fixed-cost extensive margin is non-monotone, with predicted ownership of 21.2% at the medium grid (60×20), 19.3% at the production grid (80×30), and 18.6% at the fine grid (100×40); the finest 120×50 grid sits about 0.3 pp below production. The variation is small relative to the leading Shapley contributions (pricing loads and survival pessimism) and does not threaten the top of the ranking; it is not small relative to the minor channels (Med+R-S, state-dependent utility, inflation, public-care aversion), whose attributions are read as near zero rather than as sharp points (Appendix Table A5).

3 Calibration and External Checks

Table 1 summarizes the parameter values. Each comes from a published source or is calibrated to HRS moments.

Preferences. We set $\gamma = 2.5$, within the 1–3 range Chetty (2006) estimates from labor supply elasticities. Lockwood (2012) uses $\gamma = 2$; the slightly higher value reflects the richer model environment. Section 5 reports results for $\gamma \in \{1.5, 2.0, \dots, 5.0\}$.

Age-varying consumption needs. The decline rate $\delta_c = 0.02$ matches the central estimate from Aguiar and Hurst (2013), who find nondurable consumption expenditure declining at roughly 2% per year in retirement, controlling for household composition. The rate applies uniformly across health states.

Table 1: Model Parameters

Parameter	Symbol	Value	Source
Preferences			
Risk aversion	γ	2.5	Within Chetty (2006) range (1–3)
Discount factor	β	0.97	Standard
Bequest intensity	θ	56.96	Lockwood (2012)
Bequest shifter	κ	\$272,628	Lockwood (2012); De Nardi (2004)
Demographics			
Entry age		65	Retirement
Maximum age		110	Lockwood (2012)
Risk-free rate	r	0.02	Real
Health and mortality			
Health states		3	Good, Fair, Poor
Hazard multipliers	$\mu(H,t)$	See text	HRS mortality data (see text)
Quadrature nodes		9	Gauss–Hermite
Survival pessimism	ψ	0.981 (range $[0.96, 1.0]$)	O'Dea and Sturrock (2023)
Medical expenditures			
Log mean at 65 (Fair)	μ_m	7.037	Jones et al. (2018)
Annual growth		0.065	Jones et al. (2018)
Log std dev	σ_m	1.4	Jones et al. (2018)
Consumption floor	$\underline{c}$	\$6,180	Lockwood (2012); SSI + state supplement
Pre-existing income by band	$\mathrm{SS}_{\mathrm{obs}} + \mathrm{DB}_{\mathrm{obs}}$	\$18,284/\$21,188/ \$25,924/\$26,873	RAND HRS, waves 5–9
Group-annuity MWR (extension)		0.95	TSP/TIAA-class institutional
Group access by band (extension)		39.8%/49.2%/ 59.9%/45.0%	HRS pension linkage
Annuity market			
Money's worth ratio	MWR	0.87	Mitchell et al. (1999); Wettstein et al. (2021)
Fixed cost	F	\$2,500	Author's choice; above the \$500–\$2,000 Lockwood (2012) range to include search costs
Inflation rate	π	0.02	Post-Volcker average
Preference and structural channels			
Age-varying needs decline	δ_c	0.02	Aguiar and Hurst (2013)
Health-utility weights	$\varphi(H)$	$[1.00, 0.92, 0.82]$	Finkelstein et al. (2013)
Public-care aversion	χ_{LTC}	0.49	Ameriks et al. (2020)
Exploratory behavioral channels			
SDU portfolio discount	λ_W	0.625	Blanchett and Finke (2024, 2025) (literature magnitude)
At-purchase penalty	ψ_{purchase}	0.05	Brown et al. (2008); Chalmers and Reuter (2012) (literature magnitude)

All dollar values in 2014 dollars, except the consumption floor, which retains the nominal value used by Lockwood (2012).

State-dependent utility. The health-utility weights $\varphi = [1.00, 0.92, 0.82]$ are the central midpoint of the Finkelstein et al. (2013) HRS estimates, applied through the multiplicative state-dependent-utility approach of Reichling and Smetters (2015). Good-health individuals receive full weight; Fair and Poor health reduce the marginal utility of consumption proportionally. Section 4 examines the softer translation $\varphi = [1.00, 0.95, 0.85]$ as a robustness check.

Public-care aversion ($\chi_{\mathbf{LTC}}$). The structural channel applies a utility multiplier $\chi_{\text{LTC}} = 0.49 < 1$ to consumption in the Medicaid-binding Poor-health state (when the consumption floor binds), representing the reduced utility of publicly-financed long-term care relative to self-financed care. The value is a calibration choice within the public-care-aversion evidence of Ameriks et al. (2020), expressed as a flow-utility transformation rather than a parameter they report directly. The channel is defined on Medicaid-financed care states, which exist in the model only through the medical-expenditure machinery. In counterfactuals with medical costs switched off, no publicly-financed care state exists and the transformation is inactive. The channel's direction is a priori ambiguous: aversion to the Medicaid state raises the value of retaining liquid wealth (against annuitization) but also of guaranteed income that keeps the agent off the floor (toward annuitization). In the fitted model the net effect is mildly pro-annuity, and the channel enters the Shapley decomposition with a small negative (demand-boosting) value (Section 4.4).

Exploratory behavioral parameters. The two behavioral channels are calibrated to literature magnitudes rather than estimated from US moments. SDU's $\lambda_W = 0.625$ follows the Blanchett and Finke (2024, 2025) retiree spending differential (approximately 50% from portfolio assets versus 80% from guaranteed income, implying $\lambda_W = 50/80 = 0.625$). The same calibration is applied to the Social Security claiming margin in Tharp (2026b). The at-purchase penalty $\psi_{\text{purchase}} = 0.05$ at reference consumption $c_{\text{ref}} = \$18{,}000$ is a literature-magnitude best guess consistent with cumulative-prospect-theory parameterizations (Hu and Scott, 2007) and the Brown et al. (2008) and Chalmers and Reuter (2012) narrow-framing evidence. Neither parameter is moment-matched to any US annuitization target. Section 5 reports sensitivity sweeps across $\lambda_W \in \{0.5, 0.625, 0.75, 0.85\}$ and $\psi_{\text{purchase}} \in \{0.01, 0.05, 0.09\}$.

The bequest parameters $\theta = 56.96$ and $\kappa = \$272{,}628$ are taken directly from Lockwood (2012), who estimated them from HRS bequest data at $\gamma = 2$. Using them at $\gamma = 2.5$ raises a portability concern, but a simulation-based check (Appendix C.1) shows the bequest-to-wealth ratio diverges only modestly at the baseline $\gamma = 2.5$ (about 14%), rising monotonically toward the upper end of the risk-aversion range. All results use Lockwood's original estimates, a conservative choice.

Health and mortality. Health transition matrices are calibrated to HRS panel data following De Nardi et al. (2010). The five HRS self-reported health categories are mapped to three states: Good (Excellent/Very Good), Fair (Good), and Poor (Fair/Poor). Baseline survival probabilities use a sex-blended life table: the survivorship mixture of the SSA 2003 male and female period tables at the model-eligible sample's female share (69.2%). Health-conditional survival is anchored to this table: a common age-specific factor rescales the hazard multipliers so the population-weighted survival reproduces the table at every age, preserving hazard ratios across health states. The same table prices the annuity and governs the agent's survival beliefs, so no wedge opens between prices and beliefs. The prior convention in this literature—the male SSA table from the Lockwood (2012) replication files ($q_{65} = 0.0187$, against 0.0120 for females), with unnormalized hazards—is retained as a robustness comparison: Appendix P recomputes the full nine-channel game under it (pricing loads 31.4 pp, bequests 5.8 pp, level 7.9%), and under aggregate anchoring the hazard-multiplier sweep in Section 5 holds aggregate mortality fixed while varying health-conditional dispersion by more than the male–female gap in the underlying tables, with the channel ranking intact.

Hazard multipliers are estimated from the RAND HRS longitudinal file (waves 4–16, N = 126,249 person-wave observations aged 65+). The ratios by age band are:

Age band	Good	Fair	Poor
65–74	0.49	1.00	3.29
75–84	0.60	1.00	2.77
85+	0.74	1.00	1.82

The gradient compresses at older ages, consistent with selection: survivors to 85 in Poor health have already demonstrated above-average resilience. The baseline constant multipliers $[0.50, 1.0, 3.75]$ sit at the upper end of the empirical range. The Good and Fair multipliers fall between the HRS self-reported values ($[0.57, 1.0, 2.7]$) and the ADL/IADL-based functional estimates of Reichling and Smetters (2015) ($[0.45, 1.0, 3.5]$); the Poor multiplier sits just above both, reflecting the steeper functional-limitation gradient at the younger ages where the annuitization decision is made. Section 5 reports results across the full range, including the R-S functional and HRS self-reported endpoints.

Medical expenditures. The medical expenditure process matches the moments published by Jones et al. (2018): mean out-of-pocket spending of $4,201 at age 70, growing to a combined out-of-pocket-plus-Medicaid mean of $29,703 at age 100, with a combined 95th percentile of approximately $111,509 at age 100. Health shifts both the mean and variance of log spending, with Poor health raising expected costs by a factor of roughly two relative to Fair.

Population sample and empirical targets. The wealth distribution is drawn from the RAND HRS longitudinal file, selecting single retirees aged 65–69 using filters comparable to Lockwood (2012). The sample contains 4,258 person-wave observations (waves 5–9). The main analysis restricts to the 2279 observations with liquid wealth above $5,000, as those with negligible wealth cannot meaningfully annuitize. *The empirical-target ownership rates reported below (*3.64% *lifetime contract indicator and* 6.06% *any-annuity income proxy) are computed on the same wealth-restricted subsample as the model predictions, ensuring symmetric comparison.* The structural model maps each household's state to a predicted decision rather than estimating moments from the sample distribution. The HRS observations are unweighted so that the empirical target maps directly to the cells solved in the structural decomposition.

The paper reports two HRS measures of US lifetime annuity ownership in parallel as empirical targets:

1. *Lifetime annuity contract indicator (q286 series; preferred measure).* Drawn from the HRS pension grid ("fat-file") waves 5–9. Identifies respondents reporting at least one annuity contract for which the question stem "Will this annuity continue for the rest of your life?" is answered affirmatively. In the pooled sample of 2,279 eligible person-waves, 83 report a lifetime annuity contract: 3.64% (Wilson 95% CI $[2.74\%, 4.82\%]$). Persons appear in up to five waves and the indicator is lifetime-cumulative, so both headline intervals are approximate Wilson-type intervals evaluated at the person-wave rate with a Kish effective sample size of 1263 under within-person correlation one (1,502 unique persons contribute one to five waves each; the correction is conservative for the income proxy, which is not fully time-invariant). At the person level, 75 of 1,502 sample members (4.99%) ever report a contract. The person-wave rate closely matches the 3.6% lifetime-annuity ownership rate Lockwood (2012) reports for single retirees 65–69. It is the closest available target for an SPIA-focused model, with one caveat used consistently below: the question stem identifies life-contingent payout contracts but does not distinguish individually purchased retail SPIAs from employer- or group-sourced lifetime annuities, so the measure is best read as an upper bound on retail-SPIA ownership. Section 4.10 interprets the interior-band gap accordingly.

2. *Any-annuity income proxy (`r{w}iann`; conventional measure).* The RAND HRS variable `r{w}iann` flags any positive annuity income in the wave, including DC pension withdrawals and short-period payouts not life-contingent. Pooled across waves 5–9, 6.06% of person-waves report positive annuity income (Wilson 95% CI $[4.87\%, 7.51\%]$). Much of the prior literature uses variants of this construct. It runs above the lifetime-contract rate because it conflates genuine life annuities with these non-life-contingent payouts.

Both HRS measures serve as out-of-sample comparisons for the nine-channel structural baseline (19.3%). Section 4 reports headline results against each.

Lifecycle moment check. Table A18 reports a bequest-distribution diagnostic from the lifecycle simulation: simulated vs. HRS exit-interview moments for the mean, median, and fraction with bequest greater than \$10K. These moments are not targeted in calibration. Trajectories are initialized from the weighted HRS joint distribution of wealth and health on the model's evaluation sample and decumulate against the band-level SS+DB income floor (the same population and income assignment the ownership predictions use), with no annuity purchased. The comparison serves as external discipline on bequest and decumulation behavior.

The bequest moments are partially matched in this population-initialized simulation (Table A18): the simulated median bequest sits close to the HRS exit-interview value, and the simulated mean runs above the empirical mean rather than below it. With the observed pre-existing income floor, upper-band households retain wealth to death, generating a right tail. The fraction of estates above \$10K overshoots the HRS rate: the model decumulates too little in the lower-middle of the distribution, consistent with the absence of late-life expense heterogeneity beyond medical shocks. The welfare and ownership results should not be interpreted as a full bequest-distribution fit. A full bequest-targeted calibration would change θ and κ jointly. The Lockwood (2012) estimates are kept fixed in the spirit of the unified-channel decomposition exercise.

4 Results

4.1 Sequential decomposition

Table 2 and Figure 1 present an intuitive sequential decomposition of predicted voluntary annuity ownership, adding the rational and preference channels one at a time through the nine-channel structural baseline (the exploratory behavioral extension path is tabulated in

Appendix A.8). Because this exercise is order-dependent, we use it as a descriptive path through the model and treat the exact Shapley values below as the preferred attribution. For narrative transparency the table reports medical expenditure risk and the health-mortality correlation as two separate steps; the Shapley analysis combines them into a single Med+R-S channel, with the bundling rationale in Section 4.4. This section walks the table's ordering, in which the two preference channels (state-dependent utility and age-varying consumption needs) enter before pricing loads; Section 4.2 discusses those channels' mechanisms and reports a second, rational-channels-first ordering, and Section 4.3 takes up the exploratory behavioral channels. The table covers all three blocks. The loads step below quotes the rational-channels-first path once, as an explicit illustration of the order-dependence the Shapley decomposition resolves.

Table 2: Sequential Decomposition of Predicted Voluntary Annuity Ownership

Channel	Ownership (%)	Δ (pp)	Retention	Cumulative
Frictionless population baseline	44.8	—	—	—
+ Social Security	100.0	+55.2	223.0%	2.2299
+ Bequest motives	100.0	+0.0	100.0%	2.2299
+ Medical expenditure risk (uncorrelated)	91.6	-8.4	91.6%	2.0431
+ Health-mortality correlation (R-S)	90.4	-1.2	98.7%	2.0157
+ Survival pessimism	68.8	-21.6	76.1%	1.5333
+ State-dependent utility	69.5	+0.7	101.1%	1.5499
+ Age-varying consumption needs	59.3	-10.2	85.3%	1.3219
+ Realistic pricing loads	21.3	-38.0	35.9%	0.4746
+ Inflation erosion	18.4	-2.9	86.6%	0.4110
+ Public-care aversion	19.3	+0.8	104.5%	0.4295
Observed (HRS, this sample)	3.64%–6.06%	—	—	—

Retention rate = ownership after channel / ownership before channel. Cumulative product of retention rates tracks geometric compounding.

Population benchmark with public consumption floor. The first row (no bequest motive, actuarially fair pricing, no medical expenditure risk, no inflation, and no Social Security income) predicts 44.8% ownership in the HRS population sample. This is not the theoretical Yaari full-annuitization result. The benchmark holds two US institutional features fixed across all subsequent decomposition rows: a public consumption floor (Medicaid/SSI proxy at c_{floor} = \$6,180/year) and a minimum-wealth analytical sample restriction (HRS singles with non-housing wealth ≥ \$5,000). Many low-wealth households find annuitization unattractive even in this frictionless environment because the public floor guarantees consumption at Medicaid-eligible levels, reducing the marginal value of the liquid wealth they could annuitize. This paper follows the convention of the calibrated lifecycle literature (Paschchenko 2013; Peijnenburg-Nijman-Werker 2016; De Nardi-French-Jones 2010) and bakes the safety net into the institutional environment rather than treating it as a behavioral channel.

Pre-existing annuitized income. Adding pre-existing annuitized income (Social Security plus defined-benefit pension income, COLA-protected and calibrated by wealth band) raises predicted ownership to 100.0%.[5] This income acts as a complement in the frictionless model: the guaranteed floor enables consumption smoothing and makes agents willing to lock up additional wealth in annuities. The 223.0% retention rate reflects this complementarity.

Bequest motives. Adding the DFJ luxury-good bequest specification ($\theta = 56.96$, $\kappa =$ \$272,628) leaves ownership unchanged at 100.0% (retention rate of 100%). The luxury-good $\kappa =$ \$272,628 makes marginal bequest utility nearly flat for individuals with wealth below \$200,000. In the HRS sample, median liquid wealth is approximately \$40,000.

Medical expenditure risk and health-mortality correlation (combined). Introducing stochastic medical expenditures together with the Reichling–Smetters health-mortality correlation reduces ownership to 90.4% (retention 90.4%, a -9.6 pp effect). The two are bundled into a single channel for the reason given in Section 4.4; the appendix partition check (Table A11) confirms the ranking is unchanged when they are unbundled. The mechanism: when a retiree's health deteriorates, expected remaining lifetime falls (reducing the present value of future annuity payments) and expected medical costs rise (increasing the marginal value of liquid wealth). The demand-suppressing effect depends on this correlation. Table 2 enters the two as separate sequential steps, so the medical-risk-alone and R-S-correlation-alone contributions remain visible there; the single Shapley channel bundles only their interaction. The combined channel was not jointly incorporated in the multi-channel models of Pashchenko (2013) or Peijnenburg et al. (2016).

Survival pessimism. Introducing subjective survival beliefs ($\psi = 0.981$) reduces ownership from 90.4% to 68.8% (retention 76.1%, a -21.6 pp effect). The agent overweights the probability of dying before recovering the premium.

[5] The decomposition channel labeled "Social Security" throughout (including in Tables 3 and 2) is pre-existing annuitized income, $\mathrm{SS}_{\mathrm{obs}} + \mathrm{DB}_{\mathrm{obs}}$, measured from the RAND HRS and assigned at the wealth-band level (Table 1), not per household (a household-level assignment is a natural extension); toggling the channel off zeroes both components. The label is retained for brevity because Social Security is the larger component for most single retirees. The benefit-cut counterfactual (Section 4) is the exception: it reduces the Social Security component only and holds defined-benefit pension income fixed.

Pricing loads. Repricing the annuity at MWR = 0.87 and adding a \$2,500 fixed purchase cost produces the largest single-step reduction in Table 2, entering after the two preference channels in the table's ordering. A 13% load eliminates the surplus that marginally willing buyers retained after health risk, survival pessimism, and declining consumption needs. The step also supplies the section's order-dependence illustration: on the rational-channels-first path of Section 4.2, the same repricing enters directly after survival pessimism and cuts ownership from 68.8% to 30.0% (retention 43.6%). The per-step attribution moves with the entry point; the Shapley values of Section 4.4 are the order-free attribution.

Inflation erosion. Converting the annuity from real to nominal at 2% annual inflation produces a further modest reduction (Table 2). This step is a product counterfactual. The previous steps priced a real annuity (constant purchasing power); this one switches to a nominal annuity whose real value erodes over time. Social Security income is COLA-protected and does not erode, but the private annuity payment loses purchasing power: at 2% inflation, real income from the annuity falls by 39% over 25 years.

Summary. With the six standard rational channels active and the preference channels off, the model predicts 28.2% ownership relative to a frictionless population benchmark of 44.8%; this compares channel sets and does not depend on entry order. Pricing loads and survival pessimism produce the two largest single-step reductions in the table's ordering. The standard rational literature thus substantially overshoots both empirical targets. The preference channels and the structural public-care aversion channel narrow the gap further; the exploratory behavioral channels in Section 4.3 examine it at literature magnitudes.

4.2 Preference Channel Extensions

Table A3 in the appendix tabulates the full extension path: a second, rational-channels-first ordering in which the six rational channels enter before the preference channels, with the two exploratory behavioral increments included for reference. Its per-step figures differ from those in Table 2, which orders the preference channels before pricing loads; the difference is the order-dependence the Shapley attribution resolves. The increments relevant here are quoted below.

Adding age-varying consumption needs ($\delta_c = 0.02$, calibrated to Aguiar and Hurst, 2013) reduces predicted ownership from 28.2% to 19.4%. This is the quantitatively important extension beyond the six-channel rational model: if late-life consumption needs decline, the insurance value of a level annuity stream declines as well. The resulting reduction is smaller than the survival-pessimism step but several times the inflation-erosion step.

Adding state-dependent utility ($\varphi = [1.00, 0.92, 0.82]$, the central midpoint of the Finkelstein et al., 2013 estimates) produces only a small additional reduction, from 19.4% to 18.4%. The channel's effect is insensitive to the specific mapping within the range the literature supports: the harsher FLN raw endpoint $\varphi = [1.00, 0.90, 0.75]$ yields 18.0% and the softer translation $\varphi = [1.00, 0.95, 0.85]$ used by Reichling and Smetters (2015) yields 18.7%, a spread of 0.7 pp around the production value 18.4%. In either calibration, this channel is best interpreted as a completeness check rather than a mechanism the results depend on.

The full eight-channel rational+preference model therefore predicts 18.4% ownership, a reduction from the 44.8% frictionless benchmark but, at the baseline calibration, still above both empirical targets (6.06% on the conventional any-annuity income proxy; 3.64% on the cleaner lifetime contract indicator). Like the nine-channel point below, this intermediate level inherits the model's money's-worth and risk-aversion sensitivity. Adding the structural public-care aversion channel ($\chi_{\text{LTC}} = 0.49$) yields the nine-channel structural baseline of 19.3% at the baseline calibration (its sensitivity to pricing and risk aversion is taken up in Section 4.3). The remainder of the paper uses the FLN central mapping as the production calibration.

4.3 Behavioral extensions

The nine-channel structural baseline of 19.3% sits inside a wide band: the same model predicts 15.4% at a money's-worth ratio of 0.85 and 24.3% at 0.90, and 14.4% at $\gamma = 2.4$ rising to 28.2% at $\gamma = 3.0$. Across the empirically defensible parameter ranges, the predicted level runs from below to well above the two HRS measures (3.64% on the lifetime contract indicator, 6.06% on the conventional any-annuity income proxy). Consistent with the level/ranking discipline below, the 19.3% figure is a baseline point on a steep surface, not a sharp prediction. The paper's claims rest on the channel ranking rather than the level.

An under-explored behavioral frontier. The nine-channel result reflects a scope choice rather than a judgment that behavioral factors are unimportant: the channels the annuity literature has formalized are rational and preference-based, and the headline attribution stays within them. Yet that literature has scarcely modeled behavioral frictions inside a structural annuity decomposition, and there is no a priori reason to treat them as second-order. The same channels have recently been examined on the adjacent Social Security claiming margin (where delayed claiming is effectively the purchase of additional public annuity income) and materially shift the optimal acquisition of longevity insurance there (Tharp, 2026b), which motivates examining them on the private-annuity margin. To probe that gap, we layer two behavioral channels onto the nine-channel baseline as an exploratory extension: source-dependent utility ($\lambda_W = 0.625$) and a narrow-framing at-purchase penalty ($\psi_{\text{purchase}} = 0.05$), both set to magnitudes drawn from the behavioral literature rather than moment-matched to any US annuitization target. These parameters are not identified within the model. The exercise gauges potential scale without estimating it, and the headline attribution throughout the paper holds them off.

The exercise carries one message, about scale rather than any point estimate. Source-dependent utility, on its own and at a literature-plausible magnitude, moves predicted ownership by more than any structural channel's Shapley attribution (from the nine-channel baseline of 19.3% to 56.7%), while the narrow-framing penalty alone saturates the participation margin, collapsing ownership to essentially zero. Combined, the saturation dominates, leaving the eleven-channel model at 0.1%. The source-dependent magnitude also inherits an accounting convention: medical expenses absorb income before portfolio wealth, which mechanically enlarges the SDU contribution by shrinking the full-utility income envelope in high-medical states (a portfolio-first ordering would shrink it). Because the parameters are un-identified, one saturates, and the SDU magnitude is convention-dependent, the magnitudes are not interpretable as estimates; the full sensitivity ranges, the extension-path table, and the eleven-channel Shapley all sit in Appendix A.8. The structural attribution remains informative even when un-identified channels can dominate it, because what the behavioral channels do to the nine-channel game is mechanical rather than evidential. Near saturation, coalition margins are dominated by near-zero-participation subsets, so the eleven-channel game's reordering of structural attributions reflects the saturation geometry rather than new information about the structural forces. The nine-channel game characterizes the rational and preference forces conditional on framing effects being absent; the extension quantifies what is at stake if they are not. Identifying these channels, rather than presuming them second-order, is a first-order open problem.

Out-of-sample comparison. The conventional any-annuity income proxy (6.06%, Wilson 95% CI $[4.87\%, 7.51\%]$) and the cleaner lifetime-contract indicator (3.64%, Wilson 95% CI $[2.74\%, 4.82\%]$) are reported as out-of-sample comparisons against the nine-channel baseline (19.3%). The model is disciplined by other HRS moments (the wealth distribution, health-transition and mortality-hazard rates, and Social Security and DB income profiles), but the annuity ownership rate itself is not targeted; "out-of-sample" refers to that quantity.

4.4 Exact Shapley decomposition

The sequential decomposition is order-dependent: the contribution attributed to each channel depends on which channels precede it. To obtain order-independent attributions, we compute exact Shapley values (Shapley, 1953) across two parallel cooperative games, applying the order-averaging device long used for distributional decompositions (Shorrocks, 2013) to a structural annuity model. Its known caveats travel with it: the attribution depends on the channel partition (Sastre and Trannoy, 2002), which is why Appendix F.1 recomputes the game under alternative groupings, and each channel enters as a binary on/off player evaluated across counterfactual configurations rather than at an intensity margin. One player warrants explicit unbundling: the Loads channel toggles the money's worth ratio, the fixed purchase cost, and the minimum-purchase floor together. Its attribution is read throughout as the proportional pricing wedge because the other two components are not driving it: predicted ownership is invariant to the minimum-purchase floor from \$0 to \$25,000 (Appendix F), and the zero-transaction-cost diagnostic of Section 4.9 shows the proportional load alone foreclosing the interior bands (Table 6). An eleven-player game entering the three components as separate players confirms the reading directly: the wedge alone remains the top suppressor (14.7 pp), the fixed cost is fourth (9.4 pp), and the minimum-purchase floor is negligible (2.0 pp; Appendix O). The disciplined headline attribution is the *nine-channel structural Shapley*, computed over all $2^9 = 512$ subsets of the rational, preference, and structural-LTC channels (with SDU and PED held off). An exploratory *eleven-channel extended Shapley*, computed over all $2^{11} = 2{,}048$ subsets that additionally enumerate the two behavioral parameters at their literature magnitudes, is reported separately at the end of this subsection. Medical expenditure risk and the Reichling–Smetters health-mortality correlation are combined into a single channel because the R-S mechanism's quantitative bite operates through the interaction with medical risk: without competing demand for liquid wealth in sick states, the lower expected annuity NPV does not translate into a precautionary motive against annuitization.

Nine-channel structural Shapley (headline). Table 3 reports the disciplined attribution. Two scoping notes frame the shares. First, the game's baseline already embeds the public consumption floor, held fixed across every configuration (Section 4), so the attribution divides the 25.6 pp drop from the floor-inclusive frictionless benchmark to the full model, not the larger gap to Yaari's 100%. Second, because the game contains demand-*boosting* players (Social Security, public-care aversion) whose negative values shrink the net-drop denominator, single-channel shares of the net drop read large; shares of the summed suppressing contributions are the more conservative gauge.

Pricing loads are the dominant demand suppressor on either basis, contributing 30.9 pp (121% of the net demand drop, 57% of the summed suppressing contributions). Subjective survival pessimism (12.4 pp, 48%) is the sole second-tier suppressor; the combined Med+R-

S health-mortality channel attributes near zero (0.8 pp, 3%) once aggregate mortality is anchored to the life table. Bequest motives, the single most-cited explanation for the puzzle, contribute 3.9 pp (15%), mid-pack, just behind age-varying consumption needs (5.2 pp). State-dependent utility (0.3 pp) and inflation erosion (0.2 pp) contribute small magnitudes, while public-care aversion enters with a small *negative* (demand-boosting) value (-2.0 pp), mildly raising annuitization on net.

Social Security enters with a large negative Shapley value (-26.2 pp, -102%): it is the largest-magnitude demand-boosting force in the game, raising annuitization on average rather than suppressing it, because the income floor it provides is what makes a private annuity rational. This negative sign averages over coalitions: the Shapley value credits Social Security with the complementarity it exhibits in the many frictionless coalitions where its income floor makes annuitization feasible. In the fully-loaded grand coalition that the benefit-cut counterfactual perturbs, however, the same income substitutes for private annuities (Section 4). The same duality appears, with the opposite sign, for the Med+R-S channel: its Shapley value is near zero, yet removing it from the grand coalition sharply *lowers* ownership (from 19.3% to 2.7%), so at the fully-loaded margin correlated health risk is substantially demand-boosting. Once loads have priced out the marginal buyer, high late-life medical costs are among the expenses the annuity's late-life income helps cover. That the order-independent average can mask the policy-relevant grand-coalition sign, in either direction, is itself a caution for cooperative-game attribution; the policy counterfactuals in Section 4 therefore rest on grand-coalition comparative statics rather than the Shapley values. The ranking inverts the most prominent claim in the prior literature: bequest motives, often cast as the leading explanation, are a mid-tier channel here, while pricing loads dominate. The additive Shapley averages should not obscure that the underlying channels are strong complements: the pairwise-interaction table (Appendix Table A12) shows combined effects well beyond the sum of marginals, strongest between Social Security and pricing loads (-32.6 pp). Single-lever policy counterfactuals thus understate what packages achieve (Section 4).

The nine-channel Shapley sharpens two findings from the sequential decomposition. First, pricing loads dominate demand suppression regardless of ordering, with survival pessimism as the sole second-tier suppressor and the combined Med+R-S correlation near zero once aggregate mortality is anchored. Second, the channels not jointly incorporated in the earlier multi-channel literature (the combined Med+R-S correlation, survival pessimism, and age-varying consumption needs) collectively account for a meaningful fraction of the structural demand suppression that prior decompositions omitted, even though no single one of them rivals pricing loads individually.

Table 3: Nine-Channel Structural Shapley Decomposition (Headline)

Channel	Shapley (pp)	Share (%)
Loads	+30.85	+120.6
Pessimism	+12.40	+48.5
Age needs	+5.24	+20.5
Bequests	+3.90	+15.2
Medical+R-S	+0.80	+3.1
State utility	+0.35	+1.4
Inflation	+0.22	+0.9
Public-care aversion (LTC)	-1.98	-7.7
Pre-existing income (SS+DB)	-26.20	-102.4
Total demand drop	+25.58	100.0

Exact Shapley values over all $2^9 = 512$ subsets of the nine structural channels (SDU and PED held off). Positive values are demand-suppressing contributions; negative values are demand-boosting averaged across channel configurations (pre-existing Social Security and DB income raises annuitization on average by providing the income floor, though at the fully-loaded grand coalition it substitutes for private annuities; see text).

Frictionless baseline: 44.8%. Nine-channel structural prediction: 19.3%. Total demand drop: 25.6 pp.

The eleven-channel exploratory Shapley (Table A4, appendix) layers the two behavioral channels (SDU, PED) and is reported as a sensitivity exercise rather than the disciplined attribution.

The top of the ranking is stable; the level is not. Order-independence removes one source of fragility, but the attribution must also survive the two perturbations to which calibrated annuity models are most sensitive: the risk-aversion parameter and the discrete extensive-margin statistic. It does. As γ ranges over $[2.0, 3.0]$, predicted full-model ownership moves from 0.0% to 28.2% (a knife-edge driven by the fixed-cost threshold interacting with the pricing load; the minimum-purchase floor is non-binding, leaving ownership unchanged from $0 to $25,000), yet Loads remain the top suppressor on the ownership statistic at all 5 values of γ, and the suppressor rank correlation against the continuous mean-purchase share is 0.96 at the production calibration. The claim has boundary cases, all disclosed, and the set's lower boundary is empirical rather than principled: the participation margin is degenerate (zero predicted ownership for the entire population) at $\gamma = 2.0$ as well as at $\gamma = 1.5$, but at 2.0 pricing loads still lead the suppressor ordering on the ownership statistic (26.4 versus 14.6 pp for bequests) while survival pessimism leads the continuous mean-purchase statistic, with loads ahead of bequests (15.5 versus 14.6 pp); bequests take the largest attribution on both statistics only at $\gamma = 1.5$ (25.1 versus 15.2 pp; Section 5). The plain reading is that $\gamma \in [1.5, 2.0)$ is where a bequest-first account becomes competitive; the loads-dominant claim is a claim about the range where the participation margin the paper studies is active. What moves with γ inside the set is the third slot: at low risk aversion ($\gamma \leq 2.25$) bequest motives take it, age-varying needs displace them at the baseline ($\gamma = 2.5$), and Med+R-S enters the top three only at $\gamma = 3.0$. The paper's claims are therefore the invariant dominance of pricing loads within the set, with survival pessimism second at every γ and a rotating third slot, not a fixed ordering of the full set. The level is a fragile by-product of where the population sits relative to the purchase threshold, and is best read as a range rather than a point.

Eleven-channel extended Shapley (exploratory). An eleven-channel extension additionally enumerates the two behavioral parameters at literature magnitudes (all $2^{11} = 2{,}048$ subsets). Because λ_W and ψ_{purchase} are un-identified and the at-purchase penalty saturates the model at its production value, the penalty carries the game's largest absolute Shapley contribution essentially by construction, and the behavioral players absorb attribution from the structural channels. Pricing loads fall from 30.9 to 15.5 pp yet remain the top structural suppressor, survival pessimism shrinks to 5.0 pp, and the Med+R-S channel rises to 4.2 pp, moving up the lower tiers. The behavioral players thus dilute the structural attributions without displacing loads from the top; the lower-tier shifts are a saturation artifact, since with the grand coalition near zero ownership, coalition margins are dominated by near-zero-participation subsets. The eleven-channel table and its attributions are therefore reported, with the un-identification caveat, in Appendix A.8 rather than here; the disciplined attribution is the nine-channel game above, and Section 4.3 discusses what the extension does and does not establish.

4.5 Comparison with prior work

Table A7 evaluates the channels that Pashchenko (2013) emphasized (bequest motives, minimum purchase requirements, and pricing loads) within the present framework. This is not a replication of her model, which differs in preference specification, health dynamics, and solution method. The table quantifies how much of the ownership gap those channels account for in the unified model, and how much additional reduction the broader channel set obtains. Her model predicted approximately 20% participation; the six-channel rational model here predicts 28.2%, the added preference channels bring the prediction to 18.4%, and the structural public-care aversion channel yields the nine-channel baseline of 19.3% at the baseline calibration.

Peijnenburg et al. (2016) found that full annuitization remains approximately optimal in their framework. Their model includes background risk and default risk but omits the health-mortality correlation that drives the Reichling–Smetters mechanism. Without the specific correlation between health shocks, survival, and medical costs, medical expenditure risk alone may increase annuity demand. The combined Med+R-S channel in the present model contributes 9.6 pp in the sequential decomposition but only 0.8 pp in the order-independent nine-channel Shapley attribution: the channel's bite depends on which channels are already present, which is itself evidence for the order-dependence critique.

4.6 Sensitivity to risk aversion

The decomposition results depend on calibrated parameters, particularly risk aversion γ. Table 4 reports predicted ownership across a range of γ values and inflation rates for the nine-channel structural model.

At the baseline $\gamma = 2.5$, the nine-channel structural model predicts 19.3%. Predicted ownership rises with γ: 0.0% at $\gamma = 2.0$, 14.4% at $\gamma = 2.4$, and 28.2% at $\gamma = 3.0$. Removing the structural public-care aversion channel (χ_{LTC}) gives the eight-channel 18.4% at this calibration; removing the two preference channels as well gives the higher six-channel 28.2%.

All lifecycle annuitization models share this sensitivity to γ, which reflects a purchase decision near the margin of indifference. The mortality credit is modest for healthy 65-year-olds, so the net surplus from annuitization sits near zero for many households. Small changes in risk aversion tip the optimal decision from non-purchase to purchase.

Table 4: Predicted Ownership (%) by Risk Aversion and Inflation Rate

	$\pi = 1\%$	$\pi = 2\%$	$\pi = 3\%$
$\gamma = 2.4$	15.4	14.4	12.8
$\gamma = 2.5$	20.0	19.3	18.2
$\gamma = 2.6$	22.6	22.2	21.8
$\gamma = 3.0$	28.2	28.2	28.0

Baseline: $\gamma = 2.5$, $\pi = 2\%$, DFJ bequests, MWR = 0.87, hazard multipliers [0.50, 1.0, 3.75].

A multi-gamma diagnostic decomposition (run at $\gamma = 2.0$, 2.5, and 3.0) locates where the gamma-sensitivity of the predicted *level* enters. Through the health-mortality correlation step the three decompositions stay close (90.0% to 89.2% ownership across the three values of γ), but survival pessimism splits them sharply: post-pessimism ownership is 49.5%, 68.8%, and 71.1% at $\gamma = 2.0$, 2.5, and 3.0. Pricing loads then compress all three further. The knife-edged participation level documented in Section 4 thus originates in the interaction of survival pessimism and pricing loads with risk aversion, not in the rational channels through the R-S correlation.

4.7 Joint parameter uncertainty

The single-parameter sweeps hold all other parameters at their calibrated values. As a joint stress test, we draw 1,000 parameter vectors from independent uniform distributions symmetric around the production values, spanning the Poor-health hazard multiplier, inflation, MWR, survival beliefs, and the consumption-needs slope (Appendix K details the ranges), and record the structural-baseline ownership prediction for each draw on a convergence-verified $60 \times 20 \times 51$ grid. Risk aversion is held fixed at the baseline.

The nine-channel structural prediction of 19.3% sits inside the interquartile range of the draw (median 21.6%, IQR [10.5%, 29.0%], 90% sensitivity interval [2.9%, 39.0%]). The 90% interval is wide because ownership is a strongly non-linear function of the joint draw: combinations toward the demand-supporting corners (low pricing load, weak pessimism, weak Reichling–Smetters effect) generate large positive ownership, while corners toward demand-suppression generate corner-solution zeros. Both HRS measures (3.64% on the lifetime contract indicator, 6.06% on the conventional any-annuity income proxy) are reported as out-of-sample comparisons against this range.

4.8 Ownership across the wealth distribution

The aggregate ownership prediction masks a sharp distributional structure. We partition the model-eligible sample into four fixed wealth bands by non-housing wealth ($0–30k, $30–120k, $120–350k, and above $350k). Each holds roughly a quarter of the model-eligible sample, but these are not population quartiles: the top band is roughly an eighth of all single retirees 65–69. Solving the full structural model and evaluating ownership separately within each band, predicted ownership is 0.0% in the bottom band, remains near zero through the third, then rises to 80.9% in the top band (above $350,000). The aggregate prediction of 19.3% is, almost entirely, a top-band phenomenon. This step is the proportional pricing load operating across the distribution (Figure A3 shows the underlying age-65 annuitization policy by wealth): at the baseline money's-worth ratio, immediate annuitization is value-destroying for households below the top, so the model predicts a sharp wealth threshold rather than a smooth gradient (Section 4.9 traces the band-by-band mechanism). The observed cross-section rises with wealth as the model does, but more smoothly: lifetime-annuity ownership is interior in the middle bands (Section 4.10), where the representative-band model predicts none. We read this as a scope limitation of a single-retail-SPIA model rather than a failure of the load mechanism, whose operation the positive observed wealth gradient corroborates. The interior ownership reflects better-priced employer and group annuities and within-band preference heterogeneity the model omits.

This concentration raises an identification concern: if predicted ownership lives in one band, is the channel ranking merely a description of that band? It is not. Recomputing the nine-channel Shapley separately within each band, pricing loads are the top demand suppressor in 3 of the four bands; in the top band, loads and survival pessimism are a tie at the reported precision. Survival pessimism is the runner-up in the bottom three bands. The Med+R-S correlation is second-tier only in the third band and turns demand-raising in the top band, and the pre-existing-income channel flips from a demand booster to a suppressor there (the fourth-largest in that band) as households hold enough liquid wealth that the public floor no longer drives the annuitization margin. The ranking is a property of the model rather than an artifact of where ownership concentrates.

4.9 Is the bottom-band exclusion an artifact?

The distributional structure invites a sharp objection: the bottom-band zeros, and the finding that a benefit cut leaves the most-exposed households unable to substitute, could be a corner-solution artifact of the fixed purchase cost rather than a feature of the world. We confront this with an extensive-margin gate built on the objection's own device, heterogeneous transaction costs.

For each household we compute the indifference fixed cost F^*: the SPIA transaction cost at which it is exactly indifferent between its best annuitized plan and no annuitization. A household owns under a constant cost $\bar{F}$ iff $F^* > \bar{F}$, so the structural ownership indicator is the small-dispersion limit of $\Pr(F_i < F^*)$ under any distribution of fixed costs, and the computation reproduces the structural ownership rate exactly. The diagnostic content is the *level* of F^*. In the bottom wealth band $F^* = 0$ for 100% of households, so no dispersion of transaction costs generates ownership there, and in the third band $F^* = 0$ for 77%. The zeros have two sources. Most reflect value-destruction (63% of the bottom band): already substantially annuitized through Social Security and defined-benefit income relative to their liquid wealth, and holding precautionary balances against the Medicaid consumption floor, these households place negative marginal value on a SPIA even at a zero purchase cost. The remainder (37%) hold too little wealth to meet the $10,000 minimum purchase, and so are size-excluded rather than cost-excluded. Neither group is reachable by any dispersion of transaction costs. Both interior bands carry a genuine fixed-cost-excluded group with $0 < F^*$ below the purchase cost: a 8% sliver of the second band (where 92% have $F^* = 0$) and the remainder of the third. A cost dispersion could reach these households, but under the production purchase cost both bands remain at essentially zero ownership.

The benefit-cut incidence inherits this structure, and its top-heavy concentration is robust across treatments of the participation margin. Table 5 recomputes the response to a 22% Social Security cut under four treatments of the extensive margin: the production model; a feasibility-only variant that drops the fixed cost entirely, leaving the minimum-purchase requirement and the consumption floor; a cost-smoothed variant with heterogeneous fixed costs at a fixed dispersion; and a theta-dispersed variant that assigns each household a bequest intensity from the *observed* within-band distribution of HRS bequest intentions, frozen before the cut. No band ownership rate is a calibration target in any treatment, so the exercise spends no degrees of freedom on the result. The incidence is strongly top-heavy in every treatment: the positive response concentrates in the top band, the third band adds a smaller positive response under the treatments that retain a fixed purchase cost, the feasibility-only variant reverses the third band's sign, and the bottom two bands respond by zero or slightly negatively throughout, the second band's decline reflecting lower-middle households pushed toward the Medicaid floor. A placebo lump-sum income cut of equal aggregate size produces a response that is likewise concentrated at the top, which locates the concentration in which households *can* respond (the feasibility wall of minimum purchase and consumption floor) rather than in the proportional incidence of the Social Security cut.

A channel-removal diagnostic identifies which force is responsible in the bands where the

Table 5: Social Security Cut Response: Invariance to the Extensive-Margin Treatment

			Induced response by wealth band (pp)			
Extensive-margin treatment	Base (%)	Cut (%)	<30k	30–120k	120–350k	>350k
hard	19.3	22.9	+0.0	-0.6	+1.1	+15.1
feasibility-only	30.9	30.1	+0.0	-2.9	-1.3	+1.3
cost-smoothed	19.1	22.4	+0.0	-0.8	+2.0	+12.7
theta-dispersed	21.0	24.5	+0.0	-0.9	+4.3	+11.8

Each row models the age-65 participation margin differently; all derive from the same per-household indifference fixed cost, so no band rate is a calibration target. "hard" is the production model; "feasibility-only" drops the fixed cost (minimum purchase and consumption floor remain); "cost-smoothed" applies heterogeneous fixed costs (fixed dispersion, not fitted); "theta-dispersed" applies the observed within-band bequest-intention distribution (HRS beq100), frozen before the cut. The induced response is concentrated in the top band in every treatment, with a smaller positive third-band response under the fixed-cost treatments and zero or negative responses in the bottom two bands throughout.

model most conspicuously misses the data. Table 6 recomputes the zero-cost would-annuitize share for bands 2 and 3 (the interior bands where observed lifetime-SPIA ownership sits furthest above the model's predicted zeros) under the full model and with each channel removed in turn. In band 3 fewer than a quarter of households would buy the immediate annuity even for free, yet removing the proportional pricing load lifts that share to near-universal. Removing survival pessimism alone would also lift a majority of the band, and removing the bequest motive to roughly two-thirds (Table 6); the band-3 exclusion is jointly overdetermined. The load remains the largest single contributor (well ahead of the fixed cost and of any one preference parameter) where the cross-section shows the most ownership, consistent with reading the interior gap as a product-scope limitation rather than a failure of the load mechanism.

The bottom-band exclusion is thus a structural property rather than a smoother artifact. What the single-SPIA model does not reproduce is the modest *interior* ownership the cross-section shows at low wealth (Section 4.10): observed low-wealth ownership reflects annuity products outside the immediate-SPIA frame (deferred, variable, and employer-linked products that the broad any-annuity measure captures), together with preference heterogeneity the model holds fixed, neither of which overturns the sign of the policy result. The analysis supports that sign rather than any participation level: the private annuitization a benefit cut induces is regressive, accruing to the households least exposed to the cut and bypassing those most exposed.

Table 6: Value-destruction diagnostic: would-annuitize share at zero transaction cost, by wealth band

Specification	Band 2 (\$30–120k)	Band 3 (\$120–350k)
Full structural	8.3%	23.5%
less Pricing loads (MWR)	70.5%	100.0%
less Survival pessimism	63.6%	92.4%
less Bequest	9.6%	64.3%
less Age needs	15.4%	55.2%
less Inflation	6.9%	27.3%
less Public-care (LTC)	0.3%	26.3%
less State utility	8.5%	19.7%
less Pre-existing SS+DB	0.0%	5.5%
less Med+R-S	0.0%	0.0%

Share of households in each band who would purchase the immediate annuity at a *zero* transaction cost (indifference fixed cost $F^{\star} > 0$), under the full structural model and under each one-channel-removed variant. Bands 2 and 3 are the interior bands where observed lifetime-annuity ownership most exceeds the model's zero prediction. A full-model entry near zero that jumps when a channel is removed identifies that channel as the one making annuitization value-destroying for the modal observed owner: in band 3 the removal of pricing loads moves the zero-cost share furthest, so the load, not the transaction cost, drives the predicted exclusion. Production grid.

4.10 Descriptive cross-sectional patterns

The ranking implies testable cross-sectional patterns in observed ownership, which we evaluate in HRS samples drawn under the same filters as the model's (single nonworking retirees aged 65–69, waves 5–9). The band-level ownership comparisons use the wealth-restricted analysis sample (2,279 person-waves); the multivariate logit uses the broader complete-case covariate sample (3,636 person-waves from 2,258 persons), since the covariate battery requires items unavailable for every analysis-sample observation. The resource and health channels predict that ownership rises with wealth (feasibility and the fixed-cost margin), falls with poor health at the purchase age (the Med+R-S valuation-risk channel), and falls with pre-existing annuitized income conditional on wealth (Social Security crowd-out). A weighted logit with person-clustered standard errors (Table A19) recovers all three signs, though the wealth gradient is not monotone at the very top: the highest band's coefficient sits below the third band's, sign-consistent but imprecise. Of the 7 channel-implied signs, 5 match. The matches are directional: in this modest sample the wealth-gradient and crowd-out coefficients are not individually distinguishable from zero at conventional levels. The two sign mismatches come from the expectation-based channels: subjective survival optimism (from the HRS subjective-survival probability) enters with the wrong sign, and bequest intention (the self-reported probability of leaving a $100,000 bequest) enters positively rather than negatively, and is the only precisely estimated channel coefficient. The bequest mismatch is the more informative of the two. The positive coefficient survives the direct control for housing wealth in the same regression, so it is not an omitted-housing artifact: a self-reported intention to leave a $100,000 bequest is associated with *more* annuity ownership, not less. That is the opposite of what the standard account (bequest motives as a first-order suppressor of annuitization) predicts, and it is consistent with the structural finding that bequests rank only mid-pack rather than dominant. The self-reported intention is a noisy proxy for the luxury-good bequest primitive the model uses, so the coefficient is better read as failing to support the strong-bequest prior than as a sharp rejection of any one channel. Subjective-survival reports, the other mismatch, are noisy and weakly correlated with realized mortality: the kind of expectation proxy whose identification the structural calibration is designed to bypass. The exercise is a coarse plausibility check on the resource and health channels rather than independent corroboration of the full ranking.

The same sample lets us place the model's predicted wealth profile against the observed one directly. Table 7 reports lifetime-SPIA ownership by wealth band alongside the model's band predictions. The observed profile increases across the wealth distribution (a Cochran–Armitage trend test confirms the gradient), which the model reproduces in direction. The two diverge in the interior: pooled observed ownership in the three non-top bands is statistically distinguishable from zero (the table reports a Wilson interval that excludes it), whereas the single-product model predicts none there. The displayed person-wave rates show a slight dip in the top band, but this is a pooling artifact: at the person level, where the inference is run, observed ownership is monotone increasing in wealth. The model also overshoots at the top: the threshold prediction of 80.9% sits far above the observed rate, the same level fragility documented in Section 5, and the fixed-dispersion cost smoothing barely moves it. The shape of the gradient is right, the top-band level wrong, the interior zeros contradicted. The interior gap is the product-scope limitation discussed in Section 4.9: the lifetime-contract measure still captures employer and group annuities priced better than the retail SPIA the model prices, and the representative-band agent omits within-band preference heterogeneity. This locates the interior gap in pricing access rather than in the demand structure: at retail pricing the single-product model predicts zero ownership in the interior bands, while repricing the same contract at the group level (MWR = 0.90) lifts the third band to 5.0%, roughly the observed rate, anticipating the two-product finding that access to institutional pricing generates the interior ownership. Section 4 therefore extends the model with a coverage-calibrated group product and re-runs the benefit-cut experiment inside the extended frame. We report both misses rather than fit either.

4.11 Heterogeneous welfare

Table 8 reports the consumption-equivalent variation (CEV) from annuity market access under baseline pricing, across household types defined by initial wealth, health status, and bequest motive intensity.

Closed-form compensating variation. The reported CEVs are closed-form compensating variation, defined by the consumption-scaling indifference condition $V_{\text{without}}(c \cdot (1+\lambda)) = V_{\text{with}}$. The reported cells use the nine-channel structural model, so the two exploratory behavioral channels are held off and do not enter them. Within that model, CRRA utility and the age- and health-varying consumption weights are homogeneous of degree $1-\gamma$ in self-financed consumption and so scale as $(1+\lambda)^{1-\gamma}$ under a uniform consumption scaling. Two felicity terms do not: the bequest, through the shifter κ, handled exactly below; and any period in which the Medicaid floor binds, where realized consumption is pinned at the fixed dollar floor c_{floor} (at $\chi_{\text{LTC}}\, c_{\text{floor}}$ in the Poor health state) and is invariant to λ rather than homogeneous in it. The closed form is thus exact for cells whose without-access optimum stays above the floor in every state. Every reported cell with a non-negligible compensating variation is high-wealth, where the floor binds only at advanced ages in the most adverse medical-cost realizations, states of negligible survival-weighted probability, so the

Table 7: Predicted and observed annuity ownership by wealth band

Wealth band	Observed (HRS lifetime)	Model (threshold)	Model (cost-smoothed)
\$0–30k	1.7%	0.0%	0.0%
\$30–120k	3.4%	0.6%	0.9%
\$120–350k	5.3%	0.0%	0.1%
Above \$350k	4.5%	80.9%	79.8%

Observed: unweighted lifetime-SPIA ownership in the HRS analysis sample (83 owner observations among 2279 pooled person-wave observations, from 1502 distinct persons of whom 75 ever report a lifetime annuity).

Model (threshold): the single-product structural model's predicted ownership by band. Model (cost-smoothed): the same prediction under lognormal heterogeneous transaction costs with median at the \$2,500 literature fixed cost and log-standard-deviation 0.5 — the fixed, non-fitted dispersion of Section 4.9; no band ownership rate enters its construction.

Inference is at the person level, since persons appear in up to five waves with a near-time-invariant outcome: each person counts once, with wealth band assigned at the first eligible wave and ownership if reported in any eligible wave. The observed gradient is increasing: a Cochran–Armitage trend test across the four bands gives $z = 4.44$, $p < 0.001$. Pooled person-level ownership in the three non-top bands is 4.0% (Wilson 95% CI [3.03%, 5.29%]), excluding zero, where the single-product model predicts none. The slight top-band dip in the displayed person-wave rates is a pooling artifact: at the person level ownership is monotone increasing in wealth (6.3% in the third band, 8.5% in the top), with the top-two-band difference not statistically distinguishable (Fisher exact $p = 0.30$).

Table 8: Consumption-Equivalent Variation by Household Type

Wealth	Health	No bequest	Moderate (DFJ)	Strong bequest
\$10,000	Good	0.00%	0.00%	0.00%
	Fair	0.00%	0.00%	0.00%
	Poor	0.00%	0.00%	0.00%
\$25,000	Good	0.00%	0.00%	0.00%
	Fair	0.00%	0.00%	0.00%
	Poor	0.00%	0.00%	0.00%
\$50,000	Good	0.00%	0.00%	0.00%
	Fair	0.00%	0.00%	0.00%
	Poor	0.00%	0.00%	0.00%
\$100,000	Good	0.00%	0.00%	0.00%
	Fair	0.00%	0.00%	0.00%
	Poor	0.00%	0.00%	0.00%
\$200,000	Good	0.00%	0.00%	0.00%
	Fair	0.00%	0.00%	0.00%
	Poor	0.00%	0.00%	0.00%
\$500,000	Good	1.93%	0.25%	0.00%
	Fair	1.73%	0.08%	0.00%
	Poor	0.25%	0.00%	0.00%
\$1,000,000	Good	5.95%	1.43%	0.00%
	Fair	4.68%	1.12%	0.00%
	Poor	3.21%	0.72%	0.00%

CEV: consumption-equivalent variation (%). Positive values indicate welfare gain from annuity market access. Full model with medical costs, health-mortality correlation, MWR = 0.87, inflation = 2%, $\gamma = 2.5$.
Welfare model uses representative SS income (\$23,556/yr, midpoint of the middle wealth-bin floors) for all wealth cells.

reported compensating variations are exact to the precision reported. Writing the without-access value as the sum of a consumption-felicity component and an additively separable bequest component B (bequeathed wealth is a stock, unaffected by scaling consumption along the fixed without-access saving policy), the indifference condition has the closed form $\lambda = [(V_{\text{with}} - B)/(V_{\text{without}} - B)]^{1/(1-\gamma)} - 1$. When $\theta = 0$, $B = 0$ and this reduces to the value-ratio form $\lambda = (V_{\text{with}}/V_{\text{without}})^{1/(1-\gamma)} - 1$ (under the luxury specification a bequeath-nothing optimum still carries the constant $\theta\kappa^{1-\gamma}/(1-\gamma)$, which the decomposition handles like any other bequest component); the no-bequest cells, including the largest-CEV cell, are therefore identical under the two forms, while the bequest-active cells carry the exact correction.

Under the nine-channel structural baseline, the welfare gain from annuity market access is zero for most households and positive only at high wealth with weak bequest motives, where it is largest in good health. The largest single-cell CEV is 5.9% at $1,000,000 in Good health under no bequests (Figure A4 maps the no-bequest CEV surface), falling to 1.4% under DFJ bequests at the same cell. At and below $200,000, and at Poor-health cells under strong bequests, CEV is essentially zero; the exceptions are the highest-wealth Poor-health cells under no bequests and under DFJ bequests (Table 8). At the population level, mean CEV under DFJ bequests is 0.2%, with 17.4% of households deriving any positive welfare gain and 10.0% deriving gains exceeding 1% of consumption. The population evaluation covers the 2,916 of 4,258 sampled households with positive liquid wealth below the grid ceiling, a set that does not impose the $5,000 ownership-eligibility floor; the exclusions are almost entirely zero-liquid-wealth households, for whom the access value is trivially zero, and including them at zero would lower the positive-CEV share from 17.4% to roughly 11.9%.

These small CEVs carry the same economic content as the low predicted ownership rate: most households would not optimally purchase an annuity at current prices, so market access provides little marginal value and the marginal household is approximately indifferent at the no-purchase margin. Larger welfare gains appear only when policy changes the optimal choice: lowering the load (group pricing) or modifying the choice architecture so that annuitization is the default, reported in Section 4.12.

Section 4.12 examines how supply-side reforms change these welfare results.

Normative interpretation of the behavioral channels. The exploratory behavioral channels (SDU, PED) carry a normatively contingent welfare interpretation (Bernheim and Rangel, 2009; Beshears et al., 2008): under a preference interpretation of the underlying phenomena (welfare-relevant), agents already incorporate them into their utility function and "removing" them does not increase welfare; under a bias interpretation (welfare-irrelevant), removing them captures the full rational-utility gain. The paper does not adjudicate the preference-vs-bias choice; welfare claims involving the behavioral channels carry this caveat.

4.12 Policy counterfactuals

Table 9 reports predicted ownership under twelve policy scenarios that vary pricing, inflation protection, survival beliefs, and Social Security generosity. Table A21 reports the corresponding welfare gains.

Table 9: Predicted Annuity Ownership Under Policy Counterfactuals

Policy Scenario	MWR	Inflation	Ownership (%)	Δ (pp)
Baseline	0.87	2%	19.3	—
Group pricing (MWR=0.90)	0.90	2%	24.3	+5.0
Public option (MWR=0.95)	0.95	2%	33.1	+13.8
Actuarially fair (MWR=1.0)	1.00	2%	44.3	+25.0
Real annuity, TIPS-backed	0.78	0 (real)	6.8	-12.5
Real annuity, nominal-equiv	0.87	0 (real)	20.4	+1.2
Fair + real	1.00	0 (real)	41.2	+22.0
SS cut 22%	0.87	2%	22.9	+3.6
Correct pessimism (psi=1.0)	0.87	2%	38.9	+19.6
Group + correct pessimism	0.90	2%	48.8	+29.5
Public consumption floor doubled	0.87	2%	16.1	-3.2
Best feasible package	0.90	0 (real)	48.4	+29.1
Observed (HRS, this sample)			3.64–6.06	

Baseline: $\gamma = 2.5$, $\beta = 0.97$, DFJ bequests ($\theta = 56.96$, $\kappa =$ \$272,628), $\psi = 0.960$. Population: HRS single nonworking retirees 65–69 with $W \geq$ \$5,000 ($N = 2{,}279$). Group pricing reflects TSP/employer plan MWR (James et al. 2006). SS cut: 22% reduction in Social Security benefits only; DB pension income is unaffected (projected OASI trust fund depletion in late 2032, 2026 Trustees Report).

Pricing loads as a dominant supply-side lever—for the top of the distribution. Group annuity pricing (MWR = 0.90), achievable through employer plans or government-sponsored pools (James and Song, 2006), raises predicted ownership to 24.3%. A public option at MWR = 0.95 reaches 33.1%, and an actuarially fair benchmark at MWR = 1.00 reaches 44.3%. The by-band composition of these gains matters for the distributional reading: at MWR = 0.90 predicted ownership is 96.8% in the top band but 1.1% and 5.0% in the two middle bands, and even at MWR = 0.95 the middle bands reach only 3.7% and 37.0% against 99.8% at the top. Realistic supply-side improvement therefore expands annuitization almost entirely within the top wealth band; drawing in the middle of the distribution requires pricing near actuarially fair levels, consistent with the zero-cost diagnostic of Section 4.9, where removing the load lifts the third band's would-annuitize share to near-universal.

Inflation protection. A real annuity at the same MWR (0.87) raises ownership modestly, to 20.4%, while a TIPS-backed real annuity at MWR = 0.78 falls to 6.8%, illustrating that the pricing penalty for inflation indexing offsets most of the gain from removing inflation erosion. Combining inflation protection with fair pricing reaches 41.2%.

Social Security and the incidence of a benefit cut. A 22% Social Security benefit cut raises aggregate private annuity demand to 22.9%. The cut approximates the projected shortfall in the Old-Age and Survivors Insurance (OASI) trust fund, whose reserves the Social Security Trustees project to deplete in the fourth quarter of 2032, after which roughly 78% of scheduled benefits would be payable from continuing payroll-tax revenue (Board of Trustees, Federal Old-Age and Survivors Insurance and Federal Disability Insurance Trust Funds, 2026); we implement it as a permanent reduction to the Social Security component of the income floor with defined-benefit pension income held fixed. Table A20 reports the full schedule. The aggregate figure masks a sharp incidence across the wealth distribution (Section 4.8): the response is concentrated in the top wealth band, where predicted ownership rises from 80.9% to 95.9% under the cut, with a smaller positive response in the third band, a slight decline in the second, and none in the bottom band (0.0%). The households most exposed to the cut are those for whom the pricing load makes a private annuity value-destroying even at a zero transaction cost; precautionary balances held against the Medicaid consumption floor reinforce the exclusion, and the minimum-purchase floor binds only for the minority of the bottom band below $10,000 (Section 4.9). Retail longevity insurance does not backfill the public benefit cut where it bites hardest. It expands annuitization mainly among the wealthy, who least need the substitution. The two-product extension below sharpens both halves of this statement: the response spreads only as far as the pension-linked upper-middle band, and the lower-middle band's participation falls. Whether employer- or group-sourced annuities could do better is outside the single-product model; within it, even group-level pricing reaches mainly the top band (Section 4.11), and the observed interior ownership those products appear to generate is the natural place to look. This is a stylized permanent reduction known at age 65, abstracting from transition uncertainty, announcement effects, and variation by claiming history and cohort. It also holds the money's worth ratio fixed: the influx of wealthier, healthier buyers a cut induces in the top band would in general equilibrium shift the annuitant pool and the equilibrium price, a channel the partial-equilibrium calculation does not capture.

Two-product extension: the employer channel. The single-product boundary stated in Section 4.10 invites the natural extension: give households access to the institutionally priced annuitization the observed interior ownership points to. We add a second product, the same immediate life annuity priced at MWR = 0.95 (the TSP- and TIAA-class institutional benchmark), accessible to the fraction of each wealth band with employer-pension linkage in the RAND HRS analysis sample (receipt of employer pension income: 39.8%, 49.2%, 59.9%, and 45.0% across the four bands). Access is calibrated to coverage, not to any ownership rate, preserving the paper's out-of-sample discipline, and product choice is dominance-ordered (the group product is the same contract at a better price), so band-level ownership is the access-probability mixture of the two single-product predictions, with no new state variable. Before any model output, the coverage data corroborate the employer-channel reading of the interior gradient: access peaks in the third band, where observed lifetime ownership peaks. One selection effect runs against us: access is proxied by the same pension receipt whose income level feeds the DB component of the pre-existing floor, so genuinely pension-linked households carry more pre-existing annuitization than the band average the solve assigns them; because pre-existing income substitutes for private annuities at the loaded margin, the mixture overstates group-product ownership among the truly linked. Correcting it would push predicted interior ownership lower: toward the observed rate in the third band, which the mixture overshoots, and further below it in the second.

Table 10 reports the result, and Figure 2 shows it: the single-product step function, the mixture's interior gradient, and the observed profile in Panel A; the two-signed cut response in Panel B. The mixture generates interior ownership where the single-product model predicted none: 2.14% in the second band and 22.18% in the third, against observed rates of 3.37% and 5.34%. The direction of the interior gradient is reproduced; the levels are not: the second band sits below its observed rate, while the third overshoots by roughly fourfold. The bottom band remains at zero, and the top-band overshoot persists; those misses reflect within-band preference homogeneity, not product scope.

The benefit cut in the two-product model. Re-running the 22% cut sharpens the incidence result. The response is no longer confined to the top band: the pension-linked third band responds through the group product (8.69 pp, roughly half of the aggregate response), while the top band carries 50%. But the bottom half of the wealth distribution still does not respond, and the second band's participation falls (-0.09 pp): the cut removes part of the income floor that made the group annuity rational for its marginal buyers, so the income effect

Table 10: Two-Product Extension: Group-Annuity Access, the Wealth Gradient, and the Benefit-Cut Response

Panel A: Baseline ownership by band (%)					
Band	Access	Retail	Group	Mixture	Observed
<$30k	40%	0.00	0.00	0.00	1.69
$30–120k	49%	0.64	3.69	2.14	3.37
$120–350k	60%	0.00	37.02	22.18	5.34
>$350k	45%	80.86	99.81	89.38	4.46

Panel B: 22% Social Security cut (mixture, %)				
Band	Baseline	Under cut	Response (pp)	Share
<$30k	0.00	0.00	+0.00	0%
$30–120k	2.14	2.05	-0.09	-1%
$120–350k	22.18	30.88	+8.69	51%
>$350k	89.38	97.75	+8.37	50%

Access is the fraction of each band with employer-pension linkage in the RAND HRS analysis sample (receipt of employer pension income), a coverage-calibrated proxy for access to institutionally priced annuitization; no ownership rate enters the calibration. Retail prices at MWR = 0.87; the group product at MWR = 0.95 (TSP- and TIAA-class institutional pricing). Product choice is dominance-ordered, so band ownership is the access-probability mixture of the two single-product predictions. Observed: unweighted lifetime-contract (q286) rates, person-wave, restricted sample. Panel B applies the 22% cut to the Social Security component only. A negative response is the income effect: the cut removes the income floor that made the group annuity rational for marginal lower-middle-band buyers.

dominates exactly where the substitution motive is weakest. The complement–substitute duality of Section 4.4 thus becomes cross-sectionally visible within a single counterfactual: pre-existing public income is a complement to private annuitization at modest wealth and a substitute at high wealth. Neither retail nor group markets backfill the cut for the bottom half of the distribution; giving the exposed households access to near-fair institutional pricing does not change that conclusion.

Best feasible (supply-side + information) package. For ownership counterfactuals, the "best feasible" scenario combines group pricing (MWR = 0.90), inflation protection (real annuity), and corrected survival beliefs ($\psi = 1.0$), producing 48.4% ownership at the nine-channel structural baseline. This is the upper end of the combined supply-side and information packages considered, not a realistic forecast of market participation.

Demand-side architecture. A demand-side lever distinct from pricing reform is changing the choice architecture so that annuitization is the default rather than an opt-in decision. Defaults are among the most powerful levers in retirement saving: automatic enrollment raises 401(k) participation by tens of percentage points (Madrian and Shea, 2001). Whether comparable effects extend to annuitization is an open empirical question (no large-scale annuity-default experiment exists), but by analogy demand-side architecture could rival the largest supply-side reforms considered here, while requiring no change to insurer pricing or product design.

Welfare under reform. Under baseline pricing, the welfare gain from market access is zero at and below \$200,000 across all bequest specifications and is positive only at high wealth (1.4% at \$1,000,000 in Good health with DFJ bequests, 5.9% without a bequest motive). Policy interventions that change the optimal choice raise these high-wealth gains. Group pricing (MWR = 0.90) lifts the \$1,000,000 Good/DFJ cell to 2.6%, and the "best feasible" package (group pricing, inflation protection, and corrected survival beliefs) reaches 6.4% at \$500,000 and 11.5% at \$1,000,000 (Good health, DFJ), higher still without a bequest motive (Table A21, Panel C). At and below \$200,000, CEV remains near zero even under the best-feasible package, consistent with annuitization being a high-wealth phenomenon. These welfare values are closed-form compensating variation (exact for floor-interior cells), computed as derived in Section 4.11. Each measures the value of annuity-market access *within* the stated pricing and belief environment (a partial-equilibrium access value), not the net welfare effect of enacting the corresponding reform, which would additionally reflect its fiscal cost and any equilibrium price response.

The welfare analysis reframes the annuity puzzle. For the median retiree at current market conditions, non-annuitization is the correct response to loaded pricing, nominal payment erosion, correlated health and survival risk, pre-existing Social Security coverage, and declining consumption needs. The welfare case for annuitization emerges only when policy reforms also change the optimal choice: the best-feasible package modeled here raises CEV for high-wealth households in good health from near zero to as much as 11.5% of consumption, while leaving the welfare-neutral majority of the wealth distribution unaffected.

5 Robustness

Table A9 in the appendix reports the full sensitivity analysis. Key dimensions are summarized here.

5.1 Risk aversion

Table 4 and Section 4.6 report the sensitivity of predicted ownership to γ. At the baseline $\gamma = 2.5$, the nine-channel structural model predicts 19.3% (Figure A1). Ownership is 0.0% for $\gamma \leq 2.0$, rises through 8.2% at $\gamma = 2.3$ and 14.4% at $\gamma = 2.4$, and reaches 28.2% at $\gamma = 3.0$. Removing public-care aversion gives the eight-channel 18.4%; removing the two preference channels as well gives the higher six-channel 28.2%.

Beyond the level, the attribution itself was recomputed at alternative baselines. Rerunning the full 512-subset Shapley game with the baseline money's worth ratio moved to 0.85 leaves pricing loads the top suppressor (26.0 pp, against 3.9 pp for bequests). At 0.90, loads remain the largest suppressor (20.6 pp) and the Social Security complement grows to the largest absolute value in the game. Because the predicted level is grid-sensitive, the game was also recomputed at two finer grids than production. At 100×40 the full-model level falls to 18.6% and at 120×50 to 19.0%, yet pricing loads remain the top suppressor at both (27.6 and 30.9 pp) with bequests mid-pack (3.9 and 3.9 pp). The level moves with the grid; the ranking does not. A seven-channel sub-game that drops the two multiplicative-utility channels (age-varying needs and state-dependent utility) as players, computed by restricting the 512-subset lookup, leaves loads at 30.1 pp and bequests at 3.0 pp. The attribution does not depend on the utility-weight normalization those channels introduce. At $\gamma = 1.5$, below the stability set, predicted ownership is zero for the entire population in the grand coalition, and bequests take the largest attribution (25.1 pp, against 15.2 pp for loads). We read this boundary as informative rather than damaging. When risk aversion is too low for the annuity's insurance value to overcome any friction, the attribution concentrates on the channel that forecloses ownership from the near-Yaari coalitions, and the margin the headline game ranks is degenerate there.

Inflation has competing effects in the nominal-annuity calibration: it erodes the real value of later payments but raises the initial nominal payout through the insurer's pricing, which uses the nominal discount rate. At the baseline $\gamma = 2.5$, predicted ownership falls monotonically over the empirically defensible range, from 20.0% at $\pi = 1\%$ to 18.2% at $\pi = 3\%$. Payment erosion dominates at the calibrated margin.

5.2 Health-mortality correlation

The hazard multipliers govern the strength of the Reichling–Smetters mechanism. The R-S functional estimates ($[0.45, 1.0, 3.5]$) produce 19.4% ownership; the HRS self-reported health estimates ($[0.57, 1.0, 2.7]$) produce 20.2%; the conservative specification ($[0.60, 1.0, 2.0]$) produces 20.9%. An age-varying specification that linearly interpolates the HRS estimates across the three age bands in Section 3 produces 20.8% (Figure A2), reflecting the compression of the health-mortality gradient at older ages (the Poor-health multiplier falls from 3.29 at ages 65–74 to 1.82 at 85+). Across all parameterizations, wider health gradients reduce annuity demand.

5.3 Money's worth ratio

Table 11 reports ownership under MWR values from the policy counterfactual exercise.

Table 11: Predicted Ownership by Money's Worth Ratio

MWR	Ownership (%)
0.82	9.5
0.85	15.4
0.87 (baseline)	19.3
0.90	24.3
0.95	33.1

Nine-channel structural baseline (rational + preferences + $\chi_{\text{LTC}} = 0.49$).

5.4 Survival pessimism

Predicted ownership rises monotonically across the ψ range: 19.3% at the strong-pessimism endpoint ($\psi = 0.96$), 28.1% at the baseline ($\psi = 0.981$, the per-year factor implied by the O'Dea and Sturrock (2023) 65–69 ten-year survival ratio $(0.71/0.86)^{1/10}$), and 38.9% at objective beliefs ($\psi = 1.0$). The steep sensitivity confirms survival pessimism as a quantitatively meaningful channel, consistent with its Shapley value of 12.4 pp. Across the range, the implied participation level is calibration-sensitive, but the ranking claims the paper relies on are not. Recomputing the nine-channel game at the strong-pessimism endpoint $\psi = 0.96$ leaves the ranking intact: pricing loads remain the dominant suppressor and bequest motives mid-pack (Appendix A.5). Pessimism's own contribution is larger there, as expected with more pessimistic beliefs, and it remains the second-largest suppressor at both ends of the range.

5.5 Additional robustness checks

Grid convergence. Under the production per-quartile Social Security assignment, grid refinement near the fixed-cost extensive margin is non-monotone: predicted ownership is 21.2% at 60×20, 19.3% at the production grid (80×30), 18.6% at 100×40, and 19.0% at 120×50 (Table A5), so the finest grid sits about 0.3 pp below production. The refinement path brackets the production value: the finest grid at production quadrature sits below it, while the most-refined reference with 15-node quadrature (19.8%) sits above. This is one more reason the paper stakes its claims on the ranking rather than the level. The variation across grid resolutions is small relative to the leading Shapley contributions (pricing loads and survival pessimism) and does not threaten the top ordering; it is not small relative to the minor channels (Med+R-S, state-dependent utility, inflation, public-care aversion), whose small attributions should be read as near zero rather than as precisely resolved.

Quadrature convergence. Binary ownership varies non-monotonically with quadrature order below the production rule and is stable above it (Table A5, Panel B). Values at $n_{\text{quad}} \in \{9, 11, 13, 15\}$ are 19.3%, 20.8%, 21.0%, and 20.8% at the diagnostic specification, a range of 1.8 pp. Lower-order rules ($n_{\text{quad}} \leq 7$) deviate by up to 4.4 pp because the heavy lognormal medical-expense tail interacts with the Medicaid floor. Nine-node quadrature balances accuracy and computational cost (Appendix G).

Annuitization grid. Because ownership is the discontinuous indicator $\alpha^* > 0$, the resolution of the age-65 annuitization grid could shift marginal households across the participation threshold. Holding the wealth grid and quadrature at production resolution and varying n_α over $\{51, 101, 201, 401\}$ leaves predicted ownership within a 0.4 pp band (19.0% to 19.4%, 19.3% at the production $n_\alpha = 101$; Table A5, Panel D), so the participation level is not an artifact of the annuitization-grid resolution.

Bequest specification. Under the nine-channel structural baseline, the DFJ luxury-good specification ($\theta = 56.96$, $\kappa = \$272{,}628$) produces 19.3% ownership; removing bequests raises this to 27.0%. Section 2 discusses why θ and κ must be interpreted jointly.

Behavioral parameter sensitivity. The eleven-channel extended specification is sensitive to the literature-magnitude calibrations of λ_W and ψ_{purchase}. The reported sweeps span $\lambda_W \in \{0.5, 0.625, 0.75, 0.85\}$ and $\psi_{\text{purchase}} \in \{0.01, 0.05, 0.09\}$. At the production magnitudes, the at-purchase penalty saturates participation, which is why the paper reports the nine-channel structural baseline, rather than the eleven-channel extension, as the disciplined reading.

Numerical accuracy. Euler equation residuals, grid convergence, and quadrature convergence diagnostics are reported in Appendices B, G, and N.

6 Discussion

This paper is a calibrated structural exercise, not a formal estimation. Parameters are anchored to published evidence or disclosed calibration choices. The main contribution is therefore quantitative accounting: how far the standard rational channels go, what the added preference channels contribute, and which mechanisms matter most when evaluated together.

Under the baseline calibration ($\gamma = 2.5$, MWR $= 0.87$), the six standard rational channels predict 28.2% ownership relative to a frictionless population benchmark of 44.8% (which holds the Medicaid consumption floor and the single-retiree sample restriction fixed and so sits well below Yaari's individual full-annuitization benchmark), above both empirical targets and echoing the residual overprediction in Pashchenko (2013). The two preference channels bring the prediction to 18.4%, and adding the structural public-care aversion channel yields the nine-channel structural baseline of 19.3% (6.06% any-annuity proxy, 95% CI $[4.87\%, 7.51\%]$; 3.64% lifetime-contract, 95% CI $[2.74\%, 4.82\%]$). This level is sensitive to the money's-worth ratio and risk aversion (ranging from 15.4% at MWR $= 0.85$ to 24.3% at 0.90, and from near zero at $\gamma \leq 2.2$ to 28.2% at $\gamma = 3.0$), so it brackets rather than matches the HRS targets. The disciplined reading is the channel ranking, not the level or a moment-matched fit.

The exact Shapley decomposition supplies the attribution hierarchy. In the nine-channel structural game, pricing loads are the dominant demand-suppressing contributor across the $\gamma \in [2.0, 3.0]$ stability set and at alternative baseline money's worth ratios, with survival pessimism second at every γ in the set, a third slot that rotates (bequest motives at low risk aversion, age-varying needs at the baseline, Med+R-S only at $\gamma = 3.0$), and bequest motives mid-pack at the baseline, leading only at $\gamma = 1.5$, the degenerate zero-ownership boundary. Social Security enters with a large negative Shapley value as a complement that raises demand. State-dependent utility contributes a small positive magnitude, inflation erosion is near zero, and public-care aversion enters with a small negative (demand-boosting) value in the current enumeration. That bequest motives (the literature's most-cited explanation) rank below pricing and survival beliefs corrects the received account. The exploratory eleven-channel extension (Appendix A.8) adds two behavioral channels at literature magnitudes. Both carry large absolute contributions, but the parameters are un-identified and one saturates the model, so the magnitudes indicate potential scale rather than identified attributions.

The exploratory extension carries one implication for how this class of model should be read. Channels of that size are almost never included in structural annuity decompositions (this one included until now). Because such channels are un-identified and can move predicted ownership across most of the unit interval, a structural attribution that omits them presumes they are second-order, an assumption this exercise gives no reason to make. The same dynamics rationalize a long-standing pattern in the literature. When behavioral wedges of plausible size can amplify, offset, or override otherwise clean structural variation, one should expect the dispersion in annuity-demand estimates that prior multi-channel studies display. Identifying these wedges is, on this evidence, a first-order open problem rather than a footnote.

The Social Security results illustrate the complement-substitute duality. In the frictionless environment, SS crowds *in* private annuity demand by providing an income floor that makes annuitization feasible. Once pricing loads and inflation erosion are active, the same SS income crowds *out* private annuity demand. The public-finance payoff is the incidence of a benefit cut: a 22% Social Security reduction raises private annuitization mainly in the top wealth band, with a smaller response in the third, while leaving the households the cut most exposes essentially unmoved, since for them the pricing load makes a private annuity value-destroying at any transaction cost. Private longevity insurance backfills the cut chiefly for those who least need it, while the lower-middle band's participation falls under the same cut. That asymmetry is a substantive result, not just a robustness check. It also rests on nothing cooperative-game-theoretic. The incidence is a grand-coalition comparative static of the full model whose top-heavy concentration is robust across the participation-margin treatments of Section 4.9, so a reader skeptical of Shapley attribution as an accounting device loses none of the policy conclusion.

The baseline MWR of 0.87 is drawn from Mitchell et al. (1999) and Wettstein et al. (2021).

Recent evidence suggests annuity pricing has improved, with population-table MWRs for age-65 immediate annuities now near the baseline 0.87. At MWR = 0.85, the model predicts 15.4% ownership (Table 11). Pricing remains the supply-side policy lever with the largest effect on demand in this market.

Several modeling choices condition the results.

Marital structure. The model treats all retirees as single. Kotlikoff and Spivak (1981) showed that marriage provides partial longevity insurance through intra-household risk sharing, which would further reduce annuity demand among married couples.

Housing wealth. Housing wealth is excluded. Pashchenko (2013) found that housing illiquidity reduces annuity demand by constraining the liquid wealth available for annuitization.

Riskless asset only. The non-annuitized asset is a single riskless bond at $r = 2\%$, with no equity or portfolio choice. A risky asset carrying an equity premium raises the opportunity cost of annuitization and further suppresses demand (Inkmann et al., 2011), so the omission is conservative for the loads-dominate ranking and the no-backfill incidence result, by the same direction-of-bias argument as for couples and housing.

Direction of the two largest omissions. Intra-household risk sharing and housing illiquidity both *suppress* annuity demand, so their omission is conservative for the loads-dominate ranking: a model admitting them would attribute even more of the demand gap to frictions. The same shield does not automatically cover the distributional incidence result, because housing wealth is concentrated in the middle bands where the model's interior prediction is weakest. It rests instead on the retail-SPIA scoping and the gate's margin-robustness, not on a direction-of-bias argument, and is not an artifact of the sample restriction.

Subjective survival belief structure. The survival pessimism channel uses a simple proportional scaling of one-year survival rates; the actual structure of subjective belief distortions may be more complex (O'Dea and Sturrock, 2023).

Age-varying consumption needs heterogeneity. The age-varying consumption needs channel assumes a uniform 2% annual decline; heterogeneity in this rate across health states and wealth levels is not modeled.

Exploratory behavioral parameters. The eleven-channel extended specification uses $\lambda_W = 0.625$ and $\psi_{\text{purchase}} = 0.05$ as literature-magnitude best guesses rather than moment-matched calibrations. At the chosen magnitude, the purchase penalty saturates, producing an eleven-channel prediction that overshoots in the opposite direction. The disciplined reading is the nine-channel structural baseline, not the eleven-channel specification. A future US-moment-matched calibration of the behavioral channels would require an external annuitization moment that we do not adopt, to preserve the out-of-sample reading.

The behavioral evidence base is wide. Blanchett and Finke (2024, 2025) document the spending differential that motivates the SDU λ_W parameter. Hu and Scott (2007) showed analytically that cumulative-prospect-theory loss aversion reduces optimal annuitization substantially; Brown et al. (2008) and Chalmers and Reuter (2012) document the corresponding empirical patterns. Ameriks et al. (2020) document the importance of long-term care aversion in late-life saving decisions (operationalized through χ_{LTC}). Madrian and Shea (2001) documents the dominance of defaults in retirement savings behavior more generally.

Deferred income annuities (DIAs) and qualified longevity annuity contracts (QLACs) are prominent alternatives to immediate annuities. In an extension (Appendix E), DIA ownership rates are roughly half of SPIA rates. The lower MWR on deferred products offsets the actuarial advantage of concentrating payments in late life: Wettstein et al. (2021) estimate an MWR near 0.50 for deferred annuities beginning at age 85, and we assume 0.50 for the age-80 product and 0.45 for the age-85 product, extrapolating their deferral gradient.

For policy, the counterfactual analysis identifies two levers on different margins. Supply-side reform (group pricing at MWR = 0.90) raises the value of annuitization at the structural baseline. Demand-side reform (annuitization as default) is motivated by the broader evidence on the power of defaults in retirement saving (Madrian and Shea, 2001), though no large-scale annuity-default experiment yet exists. These levers are not substitutes: pricing reform leaves the demand-side default-vs-opt-in margin unaddressed, and default architecture without pricing reform fails to capture households for whom the rational valuation gap is binding.

7 Conclusion

This paper developed a structural lifecycle decomposition of the annuity puzzle. A nine-channel rational-plus-preference baseline predicts 19.3% US voluntary annuity ownership at the baseline calibration, evaluated at the focal O'Dea–Sturrock survival-belief translation $\psi = 0.981$ (19.3% at the strong-pessimism robustness endpoint $\psi = 0.96$). That level overpredicts the two untargeted HRS measures severalfold (6.06% on the conventional any-annuity income proxy, 95% CI $[4.87\%, 7.51\%]$; 3.64% on the cleaner lifetime-contract indicator, 95% CI $[2.74\%, 4.82\%]$), sitting inside a wide money's-worth and risk-aversion band that brackets both; the paper's contribution is the order-independent attribution and the stability of the ranking, not resolution of the level. This does not imply that behavioral factors are unimportant. An exploratory extension adds two behavioral channels at literature-plausible magnitudes. Source-dependent utility alone moves predicted ownership by more than any structural channel's Shapley attribution (to 56.7%), and the narrow-framing penalty alone saturates the participation margin, collapsing ownership to essentially zero. The channels are un-identified, move ownership in opposite directions individually, and the narrow-framing penalty saturates the combined eleven-channel model (0.1%), so the exercise does not rank them or treat their magnitudes as estimated. Its message is a gap: behavioral forces of plausible size can outweigh the rational and preference channels the structural literature studies almost exclusively, yet they have gone largely unexplored in decompositions of this kind. The disciplined attribution is the nine-channel structural baseline. Whether behavioral channels are second-order is an open question this paper is built to raise rather than settle.

The nine-channel structural Shapley over all $2^9 = 512$ subsets is the paper's preferred order-independent attribution. Pricing loads are the dominant demand-suppressing contributor across the plausible risk-aversion range and at alternative baseline money's worth ratios. Survival pessimism alone occupies the second tier at the baseline; the combined Med+R-S health-mortality channel joins the top three only at $\gamma = 3.0$. Bequest motives, the literature's leading explanation, rank mid-pack at the baseline and lead only at the disclosed $\gamma = 1.5$ boundary. Social Security enters with a large negative Shapley value: averaged across channel configurations it is a complement that raises demand, even though the same income substitutes for private annuities at the fully-loaded margin that the benefit-cut counterfactual perturbs (Section 4). The exploratory eleven-channel extension (Appendix A.8) shows that adding two un-identified behavioral channels of plausible size can outweigh, rescale, and reorder the structural contributions, a reason to read structural-only attributions with caution and to expect the dispersion in annuity-demand estimates that the prior multi-channel literature displays.

Welfare and policy implications follow from the same calibration. The benefit-cut application (with the coverage-calibrated two-product extension) yields a two-signed incidence: private annuitization rises among pension-linked upper-middle and wealthy households while falling in the lower-middle band, so neither retail nor group markets backfill the cut for the bottom half of the wealth distribution. Under current pricing, the welfare gain from annuity market access alone is concentrated at high wealth (5.9% at \$1,000,000 in Good health without a bequest motive; near zero at and below \$200,000), because most calibrated households would not optimally buy. Larger welfare gains require policy reforms that change the optimal choice: the best-feasible package reaches 11.5% at \$1,000,000 in Good health, higher still without a bequest motive, while remaining near zero below the top of the wealth distribution. The package combines supply-side pricing reform (group pricing), product design (an inflation-protected real annuity), and corrected survival beliefs. Changing the choice architecture through defaults is a further demand-side lever, operating on the choice-environment margin rather than on the value of annuitization, that the present model does not quantify.

Research framing should therefore shift from "why don't retirees annuitize?" to "which channels matter most, and what mix of supply-side and demand-side interventions would make annuitization welfare-improving, and for whom?"

Data Availability

Replication code (Julia) and processed calibration data are available at https://github.com/DerekTharp/annuity-puzzle-model. Health transition matrices and wealth distributions are computed from the RAND HRS Longitudinal File (public use, available at https://hrsdata.isr.umich.edu); the two person-level intermediate extracts are not redistributed, per the HRS conditions of use, and are regenerated from the user's own RAND HRS download by the documented pipeline switch. All other calibration parameters are anchored to published sources cited in the text or disclosed in the text as calibration choices.

Declaration of Competing Interest

The author owns a financial advisory business whose revenue is predominantly asset-under-management fees. The author is licensed to sell income annuities.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies in the writing process

The author used Claude (Anthropic) to assist with improving the readability and language of the manuscript and analysis code. The author reviewed and edited all content and takes full responsibility for the published article.

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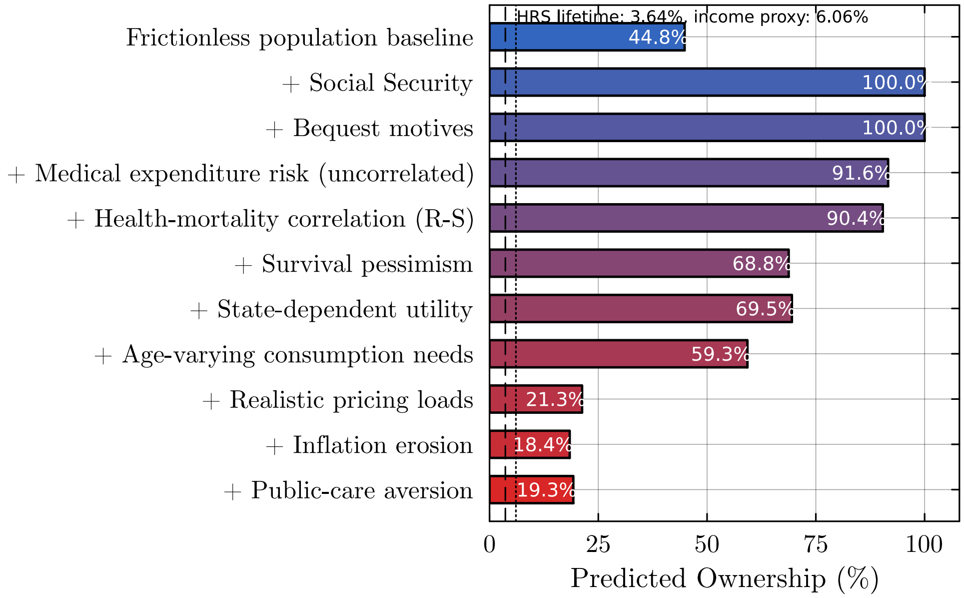

Figure 1: Sequential decomposition of predicted annuity ownership (nine-channel structural model). Each bar shows predicted ownership after adding the labeled channel to all previous channels; the final bar is the nine-channel structural baseline. This single ordering is shown for intuition only—the order-independent attribution is the Shapley decomposition (Section 4). The dashed and dotted lines mark observed ownership in the present sample (3.64% on the cleaner lifetime-contract indicator, 6.06% on the conventional any-annuity income proxy).

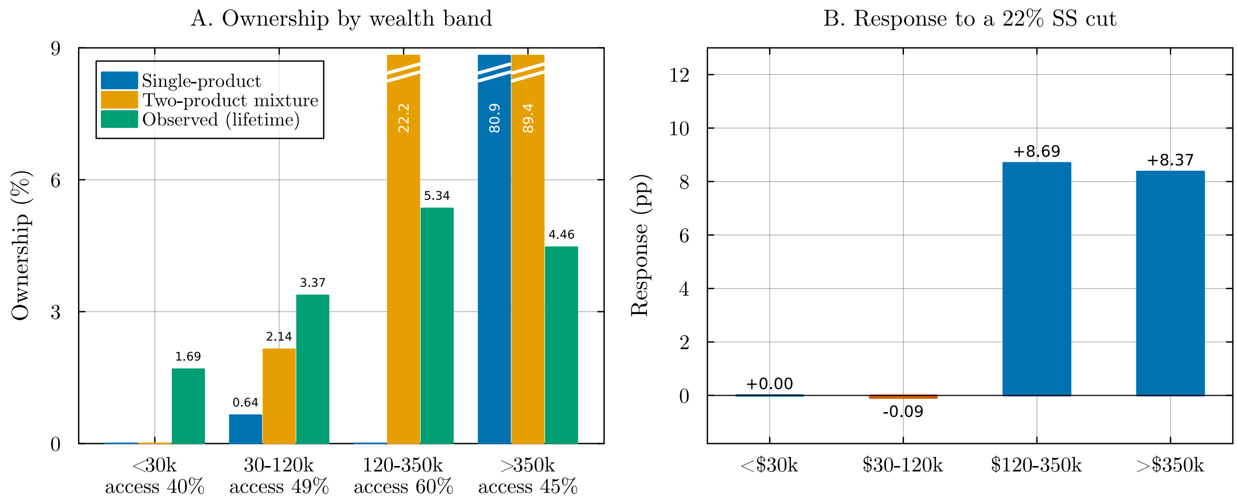

Figure 2: The two-product extension by wealth band. Panel A: predicted ownership under the single-product (retail SPIA) model, the two-product access mixture, and observed lifetime-contract ownership; group-annuity access rates (employer-pension linkage, RAND HRS) are printed beneath each band. The two top-band model bars exceed the axis limit and are truncated at the break marks; their values (80.9% and 89.4%) are printed within the bars. Panel B: the change in mixture ownership under a 22% Social Security benefit cut. The negative lower-middle response is the income effect: the cut removes part of the income floor that made the group annuity rational for marginal buyers.